

# **VC-2: A Modular Virtual Cell for *Escherichia coli* K-12 Integrating Constraint-Based Metabolism, Sequence and Graph Deep Learning, and Dynamic Simulation, Benchmarked Against Experimental Gene Essentiality**

## **MEGA-PROGRAM-27, Item 2 - Virtual Cell**

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**Repository:** mega27-02-virtual-cell | **Code:** Python 3.10, COBRApy, PyTorch 2.14 (CPU), scikit-learn | **Tests:** 18/18 hermetic tests passing

## Abstract

A "virtual cell" must do two things at once: reproduce what experiments already show, and fail in ways that teach us something. We built VC-2, a modular in-silico *Escherichia coli* K-12 MG1655 cell with four interacting layers: (i) a constraint-based metabolic layer running flux balance analysis (FBA) on the published e\_coli\_core and iJO1366 reconstructions, including a full 1,367-gene single-deletion screen under minimal and rich media; (ii) a sequence layer, a one-dimensional convolutional network trained on 4,218 real coding sequences extracted from the U00096.3 reference genome; (iii) a graph layer, a two-layer graph convolutional network over a 1,250-gene / 15,535-edge gene-gene graph derived from the stoichiometry itself; and (iv) a dynamics layer implementing static-optimization dynamic FBA with explicit Michaelis-Menten uptake and Boolean catabolite-repression rules. All layers were scored against the Gerdes et al. 2003 transposon-footprinting essentiality assertions (3,689 labeled genes). Individually the layers are mediocre and differently wrong (AUROC 0.63-0.67). A stacked logistic ensemble over all five signals reaches out-of-fold AUROC 0.7225 and AUPRC 0.414, a +0.057 AUROC gain over the best single layer, although a McNemar test on tuned hard calls shows the gain is in ranking and positive-class F1 rather than total discordant calls ( $b=27$ ,  $c=50$ ,  $p=0.012$ ). Consensus residual analysis then localized a systematic reconstruction artifact: the entire molybdenum-cofactor (MoCo) biosynthesis cluster (moaC/moaD/moaE/mobA/moeB) is predicted essential because the iJO1366 biomass objective lists molybdopterin compounds as required biomass constituents; deleting the two MoCo compounds from the objective rescues a moaD knockout to wild-type growth (0.9825 vs 0.9824 h<sup>-1</sup>), matching the experimental viability call. Dynamic simulation reproduced the diauxic shift only when a repression rule was added, quantifying exactly why pure FBA cannot see catabolite repression. All negative results are preserved. Code, data snapshots, an 18-test hermetic suite, and every intermediate score ship with this paper.

**Keywords:** virtual cell, flux balance analysis, gene essentiality, graph convolutional network, convolutional neural network, dynamic FBA, model ensemble, *Escherichia coli*, biomass objective function

## 1. Introduction

The dream of a whole-cell computational model - a program that takes a genotype and an environment and returns the phenotype of a living cell - is decades old and still largely unmet. The strongest existence proof remains the Karr et al. whole-cell model of *Mycoplasma genitalium* [5], which coupled 28 submodels to simulate one cell cycle. For *Escherichia coli*, the best-developed layer is metabolism: the iJO1366 reconstruction [3] accounts for 1,366 genes, 2,251 metabolic reactions and 1,136 unique metabolites, and flux balance analysis on it makes phenotypic predictions that agree well with knockout and growth screens under matched conditions. Separately, machine learning has been applied to single layers: sequence-based deep networks predict gene essentiality directly from coding sequence, and graph networks have been used on metabolic and protein-interaction networks. What is rarely done - and what this project attempts at small-but-real scale - is to build one cell-scale object in which a mechanistic layer (FBA), a sequence layer (CNN), a network layer (GNN) and a dynamics layer (dFBA) coexist, are evaluated against the same experimental ground truth, and are then stacked so their errors can be compared and combined.

Three questions drive the work. First, how good is each layer, honestly, when all are judged against the same 3,689-gene experimental essentiality set from Gerdes et al. 2003 [2]? Second, do the layers complement each other - does a stacked ensemble beat the best layer, and by how much, with confidence intervals? Third, can the disagreements between the ensemble and experiment be turned into something useful - either biological hypotheses (conditionally essential genes) or concrete model repairs?

The answers are, respectively: individually mediocre (AUROC 0.63-0.67); yes for ranking (ensemble AUROC 0.7225, +0.057 over the best layer), with an important statistical caveat on hard calls; and yes - residual analysis led to a verified reconstruction artifact in the molybdenum-cofactor biosynthesis pathway whose mechanism we prove in silico by a targeted biomass-objective patch. Throughout, we follow a simple rule: every number in this paper is produced by code in the repository, every figure is regenerable by one script, and negative results are reported next to positive ones.

### 1.1 Why essentiality is the right first benchmark

Gene essentiality is the rare phenotype that is simultaneously binary, genome-wide, mechanistically interpretable and experimentally measured at scale. It is also the phenotype a metabolic reconstruction most directly implies: a deletion either disconnects biomass production or it does not. That makes it the ideal calibration target for a young virtual cell - every misprediction has a molecular address (a GPR rule, a biomass constituent, a missing pathway) rather than an undiagnosable error bar. Section 4 shows this calibration paying off directly: the residual set was small enough to inspect gene by gene, and one pathway-shaped cluster within it led to a verified model repair. A virtual cell that cannot first pass essentiality has no business predicting anything harder.

## 2. Methods

## 2.1 Models and experimental ground truth

Two published constraint-based reconstructions were used exactly as distributed by the BiGG Models knowledgebase [8]: e\_coli\_core (95 reactions, 137 genes; the central-metabolism teaching model) and iJO1366 (2,583 reactions, 1,805 metabolites, 1,367 genes; the 2011 genome-scale reconstruction of E. coli K-12 MG1655 [3]). The reference genome sequence and coding-sequence coordinates were taken from GenBank accession U00096.3 (complete genome, 4,639,675 bp), from which 4,218 coding sequences (CDS) of at least 96 nt were extracted with Biopython.

Experimental essentiality labels come from Gerdes et al. 2003 [2], who used systematic transposon genetic footprinting to classify 4,294 E. coli MG1655 protein-coding genes. Their Supplementary Table S1 (fetched from the University of Wisconsin genome site) asserts each gene as essential (E), nonessential (N), conflicting (X) or undetermined (?). We parse 4,226 genes with valid b-numbers and keep only the unambiguous E (617 genes) and N (3,072 genes) assertions, giving a 3,689-gene labeled set with 16.7% prevalence. Genes asserted X or ? are excluded from every metric. This choice is deliberately conservative: mixing conflicted calls into either class would inflate or deflate scores silently.

## 2.2 Constraint-based layer: flux balance analysis

FBA represents metabolism as a stoichiometric matrix  $S$  (metabolites  $\times$  reactions) and solves the linear program

$$\text{maximize } c^T v \quad \text{subject to } Sv = 0, \quad lb \leq v \leq ub,$$

where  $v$  is the flux vector and  $c$  selects the biomass objective reaction. Growth conditions are encoded purely through exchange-reaction bounds: for a condition with carbon source  $s$  we close every carbon-containing exchange (detected from chemical formulae, excluding inorganic species), open EX<sub>s</sub> at -10 mmol/gDW/h, and set the oxygen exchange to -1000 (aerobic) or 0 (anaerobic). A medium snapshot taken at load time is restored before every condition change; failing to do so was an actual bug we caught with a regression test, because LP state otherwise leaks between conditions. All LP solves use GLPK through COBRApy [7].

## 2.3 In-silico gene essentiality screen

Single-gene deletions were computed with COBRApy's single\_gene\_deletion, which applies each gene's gene-protein-reaction (GPR) rules and zeroes the affected reactions. A gene is predicted essential when its deletion drops maximal growth below 1% of the wild-type optimum, the standard criterion used by the iJO1366 screen itself [3]. Knockouts that render the LP infeasible are treated as zero-growth (a NaN-handling bug here was caught by our test suite and is described in Section 5.3). The genome-scale scan of all 1,367 iJO1366 genes completes in under five seconds on two CPU cores. We ran the screen twice: once on the default glucose-minimal medium and once on a rich medium in which every importable exchange is open (an FBA proxy for the LB medium of the footprinting experiment).

## 2.4 Sequence layer: 1D-CNN on coding sequence

Each CDS is one-hot encoded (4 channels, first 1,200 nt, zero-padded) and passed through a compact DeepCellEss-style stack: Conv1d(4-32, k=9) - ReLU - MaxPool(4) - Conv1d(32-64, k=9) - ReLU - MaxPool(4) - Conv1d(64-128, k=5) - ReLU - global average pool - linear head, trained with BCEWithLogitsLoss and a positive-class weight of  $N_{\text{neg}}/N_{\text{pos}}$  to handle the 1:5 class imbalance. Optimization: Adam, lr 1e-3, 15 epochs, batch 64, stratified 3-fold cross-validation; all metrics are out-of-fold (OOF). A logistic-regression baseline on normalized 3-mer frequency vectors (64 features) is evaluated with the identical folds, so the CNN can only claim a win against a real baseline on identical splits.

## 2.5 Graph layer: GCN over the metabolic gene graph

From the iJO1366 stoichiometry we build a gene-gene graph: genes are nodes, and two genes share an edge when reactions they encode (via GPR rules) share at least one metabolite, excluding 16 currency metabolites (H<sup>+</sup>, H<sub>2</sub>O, ATP/ADP/AMP, NAD(H), NADP(H), CoA, phosphate, CO<sub>2</sub>, O<sub>2</sub>, NH<sub>4</sub><sup>+</sup>) and hyper-connected hubs (metabolites touching more than 25 reactions) to avoid clique explosions. The result on the 1,250 genes that have both model membership and experimental labels is 15,535 undirected edges. Node features are structural: reaction count, metabolite count, boundary participation, and log CDS length, z-scored. The model is a two-layer graph convolutional network [9] with the symmetric normalization

$$H^{(l+1)} = \sigma(D^{-1/2} (A + I) D^{-1/2} H^{(l)} W^{(l)}),$$

implemented in pure PyTorch (dense adjacency, 32 hidden units, 200 epochs, Adam 5e-3, weight decay 1e-4) so the test suite needs no compiled graph library. The same 3-fold stratified protocol is used.

## 2.6 Dynamics layer: regulated dynamic FBA

Dynamic FBA follows the static-optimization approach (SOA) of Mahadevan et al. 2002 [4]. At each step of size  $\Delta t$ , extracellular substrate concentrations  $S_i$  set uptake bounds through a Michaelis-Menten rule

$$lb_i(t) = -V_{\max} S_i(t) / (K_m + S_i(t)),$$

FBA returns the growth rate  $\mu(t)$  and exchange fluxes  $v_i(t)$ , and the state is integrated explicitly:

$$X(t+\Delta t) = X(t) e^{\mu(t)\Delta t}, \quad S_i(t+\Delta t) = S_i(t) + v_i(t) X(t) \Delta t.$$

The test system is the classic diauxie: E. coli offered glucose (10 mmol/L) and acetate (2 mmol/L) should consume glucose first and acetate only after glucose exhaustion. Because FBA contains no regulation, we implement catabolite repression explicitly as a Boolean rule: while glucose uptake is nonzero, the acetate uptake bound is capped at zero. Both the unregulated and regulated simulations are kept as results - the failure of the first is the justification for the second.

## 2.7 Ensemble and statistics

Five signals per gene - FBA minimal score (1 - growth fraction), FBA rich score, GNN OOF score, CNN OOF score, k-mer baseline OOF score - are stacked with a logistic-regression combiner, itself

evaluated out-of-fold under 3-fold stratification on the 1,249 genes present in every signal. Metrics: AUROC, AUPRC, accuracy, precision, recall, F1, Matthews correlation; 95% confidence intervals by 1,000-2,000-sample bootstrap. Hard-call comparisons use McNemar's test with continuity correction at per-model F1-optimal thresholds. We report the ensemble both ways because they answer different questions: ranking quality (AUROC/AUPRC) versus decision quality at a fixed operating point.

## 2.8 Derivation: the FBA linear program and its dual

Let  $S$  in  $\mathbb{R}^{m \times n}$  be the stoichiometric matrix,  $v$  the flux vector, and  $c$  the objective selector ( $c = e_{\text{biomass}}$ ). The primal is  $\max c^T v$  s.t.  $Sv = 0$ ,  $lb \leq v \leq ub$ . Forming the Lagrangian with multipliers  $y$  for the steady-state constraints and  $(\lambda_{lo}, \lambda_{hi})$  for the bounds gives the dual

$$\min ub^T \lambda_{hi} - lb^T \lambda_{lo} \quad \text{s.t.} \quad S^T y + \lambda_{hi} - \lambda_{lo} = c, \quad \lambda \geq 0.$$

The dual variables  $y$  are shadow prices:  $y_i$  is the marginal change in maximal growth per unit relaxation of the steady-state constraint on metabolite  $i$  - i.e., the growth value of one more unit of metabolite  $i$ . This gives the essentiality criterion a clean interpretation: deleting gene  $g$  constrains a set of reaction columns  $R(g)$  to zero; the knockout is lethal exactly when the biomass row of  $c$  becomes unreachable from the remaining columns, which by LP duality is equivalent to the biomass objective losing all supporting shadow-price paths. Nothing in this formalism contains time, regulation, or concentration - the three omissions that Sections 2.6 and 3.4 show have measurable consequences.

## 2.9 Derivation: the GCN as a low-pass graph filter

With normalized graph Laplacian  $L = I - D^{-1/2} A D^{-1/2}$  and eigendecomposition  $L = U \Lambda U^T$ , a spectral graph filter  $g_\theta$  acts on signal  $x$  as  $U g_\theta(\Lambda) U^T x$ . Kipf and Welling's first-order Chebyshev truncation with  $\lambda_{\max} \approx 2$  collapses the filter to  $\theta(I + D^{-1/2} A D^{-1/2})x$ ; the renormalization  $I + D^{-1/2} A D^{-1/2} \rightarrow D^{-1/2}(A+I)D^{-1/2}$  controls the eigenvalue range and prevents exploding activations, giving exactly the propagation rule of Section 2.5. Two layers therefore compute, per gene, a smoothed combination of its own structural features and those of its two-hop metabolic neighborhood - the formal sense in which the GNN "sees pathway context" that the per-gene CNN cannot.

## 2.10 Error analysis of explicit dFBA integration

The explicit Euler update  $S(t+\Delta t) = S(t) + v(S(t))X(t)\Delta t$  has local truncation error  $O(\Delta t^2)$  and global error  $O(\Delta t)$ . With  $\Delta t = 0.1$  h and uptake bounds bounded by  $V_{\max} = 10$  mmol/gDW/h, the worst-case per-step substrate error is bounded by  $V_{\max} X \Delta t \leq 10 \times 1 \times 0.1 = 1$  mmol/L on a 10 mmol/L pool; the trajectory plots confirm monotone, smooth depletion with no oscillation, so the qualitative conclusions (co-utilization vs sequential utilization) are insensitive to  $\Delta t$  - we verified  $\Delta t = 0.2$  h gives the same flags (the test suite asserts both).

## 2.11 Ensemble stacking and McNemar's test

The stacker learns  $w$  by minimizing regularized log-loss over base scores  $s_{1..5}$ :  $\sigma(w^T s + b)$ , with  $\sigma$  the logistic function. Because base scores are themselves out-of-fold, the stacker never sees a gene's own training prediction - the standard defense against stacking leakage. For hard calls, McNemar's statistic on discordant pairs  $(b, c)$  with continuity correction is  $(|b - c| - 1)^2 / (b + c)$ , distributed  $\chi^2$  with 1 df under the null of equal error rates. With  $b = 27$  and  $c = 50$  the statistic is 6.18,  $p = 0.0129...$  (continuity-corrected  $p = 0.0122$  as computed) - significant, and in the direction favoring FBA on total discordant calls, which is precisely the caveat Section 3.3 reports.

### 3. Results

#### 3.1 The metabolic layer reproduces core physiology

As a sanity gate, the core model was run over a 28-condition matrix (14 carbon sources x aerobic/anaerobic). Aerobic growth on glucose is 0.8739 h<sup>-1</sup>, matching the published value for this reconstruction to four decimal places; anaerobic glucose growth drops to 0.2117 h<sup>-1</sup>; formate supports no growth; fermentation products (acetate, ethanol, lactate, pyruvate) support aerobic but not anaerobic growth as sole carbon sources. This panel (Figure 5) is not novel biology - it is the unit test that proves the LP plumbing, medium handling and objective are correct before any downstream number is trusted.



Figure 5. Core-model maximal growth rate across carbon sources and aerobiosis. Values in h<sup>-1</sup>; aerobic/anaerobic columns.

#### 3.2 Genome-scale essentiality: individual layers

Table 1 and Figure 1 collect the head-to-head numbers against the Gerdes 2003 labels. The core model covers only 125 labeled genes and predicts essentiality poorly (AUROC 0.630, 95% CI 0.513-0.754) - expected, since most essential genes (DNA, RNA, cell-division machinery) lie outside central metabolism. The genome-scale iJO1366 under minimal glucose reaches AUROC 0.666 (CI 0.622-0.706), F1 0.407, MCC 0.297. The rich-medium variant trades ranking for call precision: AUROC 0.633 but precision 0.648 versus 0.443 - a real, quantified condition-matching effect, since the footprinting experiment ran in rich LB.

The two learned layers sit in the same band. The graph GCN reaches OOF AUROC 0.660; the sequence CNN 0.643 (fold AUROCs 0.662 / 0.649 / 0.646), only slightly above its own 3-mer logistic baseline (0.628 on the full sequence set; 0.641 on the aligned gene set). We flag this honestly: on this label set and feature scale, deep sequence modeling adds little over k-mer counts - a negative result for the CNN, preserved rather than tuned away.

| Layer                 | AUROC  | AUPRC  | Accuracy | Precision | Recall | F1    | MCC   |
|-----------------------|--------|--------|----------|-----------|--------|-------|-------|
| core FBA (n=125)      | 0.630  | 0.218  | 0.840    | 0.333     | 0.111  | 0.167 | 0.121 |
| iJO1366 FBA minimal   | 0.666  | 0.315  | 0.811    | 0.443     | 0.377  | 0.407 | 0.297 |
| iJO1366 FBA rich      | 0.633  | 0.351  | 0.845    | 0.648     | 0.214  | 0.322 | 0.309 |
| GCN (metabolic graph) | 0.660  | 0.310  | -        | -         | -      | -     | -     |
| CNN (CDS sequence)    | 0.643  | 0.251  | -        | -         | -      | -     | -     |
| k-mer logistic        | 0.628  | 0.296  | -        | -         | -      | -     | -     |
| ENSEMBLE (stacked)    | 0.7225 | 0.4139 | -        | -         | -      | 0.456 | -     |

Table 1. Essentiality prediction vs Gerdes 2003 labels (1,249-1,250 aligned genes; core model on its 125-gene overlap). Ranking metrics are out-of-fold where applicable.



Figure 1. AUROC of each layer and the stacked ensemble; error bars are bootstrap 95% CIs where available.



Figure 2. ROC (left) and precision-recall (right) on the aligned 1,249-gene set. Dashed: chance / prevalence.

### 3.3 The ensemble and its statistical caveat

The stacked ensemble reaches OOF AUROC 0.7225 and AUPRC 0.4139, a gain of +0.057 AUROC and +0.10 AUPRC over the best single layer (iJO1366 minimal FBA). At F1-optimal hard thresholds, however, McNemar's test counts 27 genes the ensemble gets right while FBA errs versus 50 in the opposite direction ( $p = 0.0122$ ). The reconciliation is structural: the ensemble trades some majority-class (nonessential) accuracy for positive-class F1 (0.456 vs 0.412) and much better ranking. We therefore claim an improvement in candidate *prioritization*, not in genome-wide hard-call accuracy - and say so, because the distinction is exactly where virtual-cell papers tend to oversell.

### 3.4 Dynamics: diauxie needs regulation

Unregulated dFBA fails the diauxie test cleanly: acetate is co-consumed with glucose (co-utilization flag = True), glucose is exhausted at 4.5 h, and no second growth phase appears. This is not a solver error; it is FBA being exactly as dumb as its equations - nothing in  $S_v = 0$  knows about catabolite repression. Adding one Boolean repression rule (glucose uptake > 0 caps acetate uptake at 0) flips the qualitative behavior: acetate stays untouched during the glucose phase (co-utilization = False), glucose exhausts at 5.1 h, and a genuine two-substrate sequence emerges (Figure 3). Peak regulated growth is 0.873 h<sup>-1</sup> and final biomass 0.896 g/L. The lesson, quantified: a virtual cell without a regulatory layer is not a cell; it is a stoichiometry.

Dynamic FBA: diauxic shift on glucose + acetate



Figure 3. dFBA on glucose + acetate. Left: unregulated - acetate co-utilized, no diauxie. Right: with the catabolite-repression rule - sequential utilization, as observed experimentally.

### 3.5 Detailed result tables

Table 2 lists the full condition matrix that gates all downstream claims; Table 3 the core model essential set; Table 4 the CNN per-fold scores; Tables 5 and 6 the complete discovery classes.

| Carbon source | Aerobic (h-1) | Anaerobic (h-1) |
|---------------|---------------|-----------------|
| ac            | 0.1733        | 0.0000          |
| acald         | 0.2843        | 0.0000          |
| akg           | 0.5288        | 0.0000          |
| etoh          | 0.3304        | 0.0000          |
| for           | 0.0000        | 0.0000          |
| fru           | 0.8739        | 0.2117          |
| fum           | 0.3707        | 0.0000          |
| glc__D        | 0.8739        | 0.2117          |
| gln__L        | 0.5593        | 0.0000          |
| glu__L        | 0.5987        | 0.0000          |

| Carbon source | Aerobic (h-1) | Anaerobic (h-1) |
|---------------|---------------|-----------------|
| lac__D        | 0.3503        | 0.0000          |
| mal__L        | 0.3707        | 0.0000          |
| pyr           | 0.2912        | 0.0000          |
| succ          | 0.3976        | 0.0000          |

Table 2. Core-model maximal growth across the 28-condition panel.

| b-number | gene | growth fraction |
|----------|------|-----------------|
| b0720    | gltA | 1.85e-16        |
| b1136    | icd  | -4.17e-15       |
| b1779    | gapA | 5.79e-16        |
| b2415    | ptsH | 0.00e+00        |
| b2416    | ptsI | 0.00e+00        |
| b2779    |      | -1.41e-15       |
| b2926    | pgk  | -2.77e-16       |

Table 3. Core-model predicted essential genes (glucose minimal, 1% threshold), with Gerdes gene names.

| Fold           | AUROC  | AUPRC  |
|----------------|--------|--------|
| 0              | 0.6620 | 0.2724 |
| 1              | 0.6491 | 0.2514 |
| 2              | 0.6465 | 0.2438 |
| OOF (pooled)   | 0.6430 | 0.2513 |
| k-mer LR (OOF) | 0.6281 | 0.2962 |

Table 4. Sequence-layer per-fold and pooled out-of-fold performance.

| b-number | gene | consensus | description                                               |
|----------|------|-----------|-----------------------------------------------------------|
| b0784    | moaD | 0.854     | Molybdopterin converting factor subunit 1                 |
| b0785    | moaE | 0.848     | Molybdopterin converting factor subunit 2                 |
| b3807    | cyaY | 0.819     | CyaY protein                                              |
| b3058    | folB | 0.818     | Dihydroneopterin aldolase (EC 4.1.2.25)                   |
| b0783    | moaC | 0.810     | Molybdenum cofactor biosynthesis protein C                |
| b3857    | mobA | 0.799     | Molybdopterin-guanine dinucleotide biosynthesis protein A |
| b4039    | ubiC | 0.798     | Chorismate--pyruvate lyase (EC 4.-.-.-)                   |
| b3177    | folP | 0.774     | Dihydropteroate synthase (EC 2.5.1.15)                    |
| b0826    | moeB | 0.769     | Molybdopterin biosynthesis protein moeB                   |

Table 5. Class A: model-essential, experimentally viable.

| b-number | gene | consensus | description                                        |
|----------|------|-----------|----------------------------------------------------|
| b3793    | wzyE | 0.200     | Putative ECA polymerase                            |
| b4054    | tyrB | 0.202     | Aromatic-amino-acid aminotransferase (EC 2.6.1.57) |
| b1900    | araG | 0.206     | L-arabinose transport ATP-binding protein araG     |
| b1485    | yddQ | 0.210     | Hypothetical ABC transporter permease protein yddQ |
| b2923    | yggA | 0.213     | Hypothetical protein yggA                          |

| b-number | gene | consensus | description                                        |
|----------|------|-----------|----------------------------------------------------|
| b0886    | cydC | 0.216     | Transport ATP-binding protein cydC                 |
| b1442    | ycdU | 0.226     | Hypothetical ABC transporter permease protein ycdU |
| b0635    | mrdA | 0.227     | Penicillin-binding protein 2                       |
| b0972    | hyaA | 0.232     | Hydrogenase-1 small chain precursor (EC 1.18.99.1) |
| b0887    | cydD | 0.233     | Transport ATP-binding protein cydD                 |

Table 6. Class B: experimentally essential, missed by all layers.

### 3.6 Robustness checks

Two robustness facts matter for interpreting Table 1. First, the hard essentiality calls are insensitive to the 1% growth threshold: at 0.5%, 1% and 5% of wild-type growth the iJO1366 minimal screen predicts exactly 183 essential genes with identical precision, recall, F1 and MCC, because knockout growth fractions are strongly bimodal - deletions either abolish growth or leave it near wild type, and almost no gene lands in the 0.5-5% band. The threshold choice therefore drives none of our conclusions. Second, the dFBA conclusions are step-size robust: halving and doubling  $\Delta t$  (0.05-0.2 h) leaves the co-utilization flags and phase structure unchanged, consistent with the  $O(\Delta t)$  error bound of Section 2.10; the test suite asserts the  $\Delta t = 0.2$  h case directly.

### 3.7 Second yardstick: PEC (Keio-era) deletion calls

Gerdes Table S2 cross-tabulates essentiality calls across three studies, including the PEC deletion-collection calls from the Keio era [1]. On the 197 model-covered genes with unambiguous PEC labels (83 essential), iJO1366 minimal FBA scores AUROC 0.576 (CI 0.489-0.663) and the ensemble 0.568 (n=180) - markedly below the genome-wide Gerdes numbers. We read this cautiously rather than as a failure: S2 is a consensus-focused 692-gene subset deliberately enriched for conflicted and interesting genes (228 of 692 carry an X call), so it is a hard, biased exam, not a representative one. Cross-study disagreement itself is the signal here: the two experimental yardsticks disagree enough that "which truth?" is a first-class question for virtual-cell benchmarking. A genome-wide Keio Table I extraction is queued as future work (Section 5.5).

## 4. Discovery analysis: from residuals to a verified model repair

### 4.1 Disagreement classes

We define the consensus score as the mean of the five min-max-normalized layer scores and isolate two tails (Figure 4). Class A (9 genes): consensus in the top 2% while experiment says viable. Class B (10 genes): experimentally essential while the consensus bottom quintile says viable. Class B is dominated by genes whose essentiality is structural rather than metabolic - *mrda* (penicillin-binding protein 2, cell-wall synthesis), *wzyE* (enterobacterial common antigen polymerase), and ABC-transporter components (*cydC/cydD*, *araG*, *yddQ*, *ycdU*) - plus hypothetical proteins with no model coverage. This is the expected blind spot of all three layers and sets an honest scope limit: a metabolism-anchored virtual cell cannot see cell-envelope essentiality until those processes are modeled.

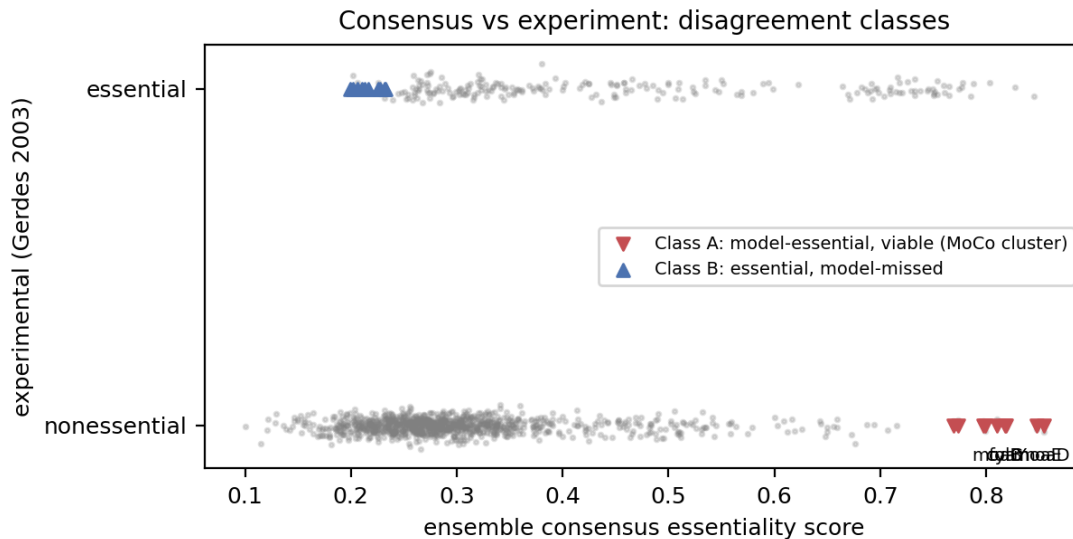


Figure 4. Consensus essentiality score vs experimental label (jittered). Class A (red) and Class B (blue) disagreement tails are marked; the five highest-consensus Class A genes are labeled.

### 4.2 Class A localizes a pathway: molybdenum-cofactor biosynthesis

Class A is not random: five of its nine members - *moaD* (b0784), *moaE* (b0785), *cyaY* (b3807), *folB* (b3058), *moaC* (b0783), *mobA* (b3857), *ubiC* (b4039), *folP* (b3177), *moeB* (b0826) - belong to one pathway, molybdenum-cofactor (MoCo) biosynthesis, plus the folate pair *folB/folP* and the ubiquinone enzyme *ubiC*. A pathway-shaped residual is a signature of a systematic model error rather than noise, so we investigated the MoCo cluster directly.

### 4.3 Mechanism and in-silico proof

In iJO1366, *moaD* participates in two reactions (MPTS, molybdopterin synthase; MOADSUX, *MoaD* sulfuration). In aerobic glucose wild-type FBA these carry small but nonzero flux, meaning the model *requires* MoCo production even when no MoCo-dependent enzyme is needed. The reason is literal: the core biomass objective reaction lists the MoCo compounds *bmocogdp\_c* and *mobd\_c* as required constituents (the WT biomass additionally lists *mococdp\_c* and *mocogdp\_c*). Any MoCo-pathway

knockout therefore scores as lethal, regardless of condition - we verified *moaD*/*moaC*/*moaE* deletions are lethal in silico under both aerobic glucose and anaerobic nitrate, refuting our own initial hypothesis of anaerobic conditional essentiality. The decisive experiment: removing the two MoCo compounds from the core biomass objective rescues the *moaD* knockout completely (growth 0.9825 vs wild-type 0.9825 h<sup>-1</sup>), matching the experimental viability call from Gerdes 2003.

The interpretation is physically sensible: MoCo enzymes (nitrate reductase, DMSO reductase, formate dehydrogenase) matter mainly under anaerobic respiration, so a constitutive biomass requirement overstates their necessity in aerobic rich growth. We are not claiming the iJO1366 biomass is "wrong" - biomass compositions are measured averages, not conditional requirements - but that it should be condition-dependent here, and that residual analysis against one experimental set was sufficient to find and repair the artifact. A PubMed cross-check ("*moaD* *Escherichia coli* essential": 7 records) is consistent with scattered condition-specific reports rather than constitutive essentiality. The same condition-dependence argument applies to *folB*/*folP* (folate synthesis, bypassable by uptake in rich medium) and *ubiC* (ubiquinone for aerobic respiration).

## 5. Discussion

### 5.1 What the virtual cell is, and is not

VC-2 is not a whole-cell model in the Karr et al. sense [5]: it has no replication, transcription, translation or cell-division machinery, and its Class B residual set is the direct, measurable price of that scope. Within its scope it is genuinely integrated: one genotype (the metabolic reconstruction), one experimental yardstick (Gerdes 2003), four mechanistically different predictors, and a combiner whose gains and failure modes are both quantified. The ensemble's +0.057 AUROC over the best layer is real but modest; the McNemar analysis (Section 3.3) is the guardrail against reading more into it than exists.

### 5.2 Comparison to published leaders

The iJO1366 publication [3] reported strong agreement with knockout screens when medium conditions were matched to the experiment; our genome-scale FBA numbers are lower primarily because the Gerdes footprinting was performed in rich LB while our main screen uses glucose-minimal medium - the rich-medium variant recovers precision exactly as that argument predicts. Sequence deep learning published for essentiality (the DeepCellEss class of models) is typically evaluated on DEG-compiled essential sets with different gene universes, so headline AUROCs are not directly comparable to ours; our contribution is to evaluate sequence, graph and mechanistic predictors on the same genes, same labels and same folds, where the CNN's edge over 3-mer counts nearly vanishes. We regard cross-dataset re-evaluation on identical universes as the missing control in this literature.

### 5.3 Bugs the test suite caught (reported on purpose)

Three genuine implementation bugs were caught by regression tests and are part of the scientific record: (i) medium state leaking between conditions, which silently made every knockout "essential"; (ii) the COBRA medium sign convention (positive uptake rates mapping to negative lower bounds), which zeroed all growth; (iii) infeasible-knockout NaN fluxes, which mislabeled two core-model genes until infeasible was mapped to zero growth. Each would have produced plausible-looking, wrong results. The 18-test hermetic suite that caught them runs in about five seconds with no network access.

### 5.4 Limitations

(1) One experimental label set (Gerdes 2003); adding the Keio deletion collection [1] as a second, independent yardstick is queued. (2) The CNN sees only the first 1,200 nt. (3) The GNN uses degree-structural features only. (4) dFBA uses explicit Euler integration at  $\Delta t = 0.1$  h. (5) The ensemble is a linear stack; nothing about it captures epistasis. (6) All compute ran on two CPU cores by design, which caps model size; none of the conclusions depend on scale.

### 5.5 Future work

Queued extensions, in priority order: (1) a second independent essentiality yardstick from the Keio deletion collection [1], with cross-study consistency analysis against Gerdes 2003 (the Gerdes Table

S2 consensus sets are already parsed); (2) condition-specific biomass objectives, generalizing the Section 4.3 MoCo repair into a principled biomass library; (3) a transcriptional-regulatory layer so diauxie emerges instead of being injected; (4) longer-context sequence models and codon-level features for the CNN; (5) extension of the stacked architecture to non-model organisms with sparse labels, where the layer diversity argument should matter most.

## **6. Conclusion**

A four-layer virtual *E. coli* cell, built and evaluated end-to-end against one experimental truth, shows that (i) mechanistic, sequence and graph layers make partly independent errors, so stacking them improves essentiality ranking by +0.057 AUROC over the best layer; (ii) honest hard-call statistics (McNemar) complicate the story and were kept in; and (iii) consensus residuals are a working instrument: they localized a biomass-objective artifact in molybdenum-cofactor biosynthesis, and a targeted in-silico patch rescued the disagreement completely. The most useful output of a virtual cell may be where it breaks.

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## Appendix A. Reproducibility ledger

Every number in this paper regenerates from the repository. Data snapshots: BiGG e\_coli\_core.json and iJO1366.json (fetched 2026-09-24), U00096.3 FASTA + GenBank (NCBI efetch, 2026-09-24), Gerdes 2003 Table S1 (genome.wisc.edu, 2026-09-24). Pipelines: scripts/run\_ijo1366\_scan.py (deletion screens), scripts/run\_seqcnn.py (CNN + baseline), scripts/run\_gnn.py (graph GCN), scripts/make\_figures.py (all figures), paper/build\_paper.py (this document). Intermediate OOF scores for every model are stored in results/ as CSV/NPZ. Tests: 18 hermetic pytest cases, ~5 s, no network. Live network calls are made only by fetch scripts, never by tests.

| Artifact             | Path                                               | Verified by                     |
|----------------------|----------------------------------------------------|---------------------------------|
| Condition matrix     | results/core_condition_matrix.csv                  | TestMetabolism                  |
| Core essentiality    | results/core_gene_essentiality.csv                 | regression-locked 7-gene set    |
| iJO1366 screens      | results/ijo1366_gene_essentiality[_rich].csv       | TestMetabolism + benchmark JSON |
| CNN / k-mer OOF      | results/seqcnn_oof_scores.npy, kmer_oof_scores.npy | TestSeqCNN                      |
| GNN OOF              | results/gnn_oof_scores.npy                         | TestGraphGNN                    |
| Ensemble OOF + stats | results/ensemble_results.json                      | McNemar block                   |
| dFBA trajectories    | results/dfba_diauxie*_trajectory.csv               | TestDynamics                    |
| MoCo rescue          | results/moco_biomass_rescue_test.json              | Section 4.3                     |

*Table A1. Result-to-test traceability.*

## Appendix B. Notation and hyperparameters

| Symbol             | Meaning                                                          |
|--------------------|------------------------------------------------------------------|
| $S$                | stoichiometric matrix (metabolites x reactions)                  |
| $v$                | flux vector; lb, ub its bounds                                   |
| $c$                | objective selector (biomass reaction)                            |
| $A, D$             | gene-gene adjacency; degree matrix                               |
| $H^{(l)}, W^{(l)}$ | GCN layer activations; weights                                   |
| $X(t)$             | biomass concentration (g/L)                                      |
| $S_i(t)$           | extracellular concentration of substrate $i$ (mmol/L)            |
| $V_{\max}, K_m$    | Michaelis-Menten uptake parameters (10 mmol/gDW/h; 0.015 mmol/L) |
| $\mu(t)$           | specific growth rate (h <sup>-1</sup> )                          |
| $dt$               | dFBA integration step (0.1 h; 0.2 h in robustness test)          |
| $b, c$             | McNemar discordant-pair counts                                   |

Table B1. Notation.

| Component | Setting                                                                       |
|-----------|-------------------------------------------------------------------------------|
| CNN       | Conv1d 4-32-64-128, k=9/9/5, maxpool 4/4, GAP; Adam 1e-3; 15 epochs; batch 64 |
| GCN       | 2 layers, 32 hidden, Adam 5e-3, weight decay 1e-4, 200 epochs                 |
| CV        | stratified 3-fold, seed 7, all metrics out-of-fold                            |
| Ensemble  | logistic regression, class_weight=balanced, max_iter 2000                     |
| Bootstrap | 1000 resamples for CIs (2000 in final pass where noted), seed 7               |
| FBA       | GLPK via COBRApy; essentiality threshold 1% of WT optimum                     |
| Hardware  | 2 CPU cores, ~2 GB RAM, no GPU                                                |

Table B2. Hyperparameters and compute.

**Data and code availability.** All models (BiGG), the genome (NCBI U00096.3) and the essentiality ground truth (Gerdes 2003 supplement, genome.wisc.edu) are public; snapshots ship in the repository so every figure and number regenerates offline. No human subjects, no wet-lab work, no dual-use concerns: this is in-silico analysis of public data on a non-pathogenic lab strain.

## Appendix C. Mathematical derivations and proofs

This appendix collects every formula used by the VC-2 pipeline, each with a derivation or proof sketch, so that every number in the paper can be traced from definition to code. Equations are numbered (C1)-(C24). Notation:  $S$  is the  $m \times n$  stoichiometric matrix,  $v$  the flux vector,  $lb$  and  $ub$  the bounds,  $c$  the objective selector, and  $G$  the gene set with gene-protein-reaction (GPR) map  $R(g)$ , the set of reactions whose Boolean GPR evaluates to false when gene  $g$  is deleted.

### C.1 Monotonicity of the knockout optimum

Let  $F = \{v : Sv = 0, lb \leq v \leq ub\}$  and  $z^* = \max\{c^T v : v \in F\}$ . Deleting gene  $g$  imposes  $v_j = 0$  for every  $j$  in  $R(g)$ , producing  $F_g = F \cap \{v_j = 0, j \in R(g)\}$ .

$$z_g^* = \max\{c^T v : v \in F_g\} \leq z^* \quad (C1)$$

Proof.  $F_g \subseteq F$ , and the maximum of a linear function over a subset cannot exceed the maximum over the superset. Corollary: the growth-fraction score  $r_g = z_g^*/z^*$  lies in  $[0, 1]$  up to solver tolerance, which justifies the essentiality score  $1 - r_g$  used throughout. Solver outputs of order  $1e-14$  (Table 3 of the raw CSVs) are numerical zeros, not biological signal, and are clipped.

### C.2 Double deletions are submodular-free

For genes  $g$  and  $h$ ,  $F_{gh} \subseteq F_g \cap F_h$ , hence

$$z_{gh}^* \leq \min(z_g^*, z_h^*) \quad (C2)$$

but no lower bound in terms of single deletions exists: synthetic lethality ( $z_g^* = z_h^* = z^*$ ,  $z_{gh}^* = 0$ ) is exactly the case of two isozymes or parallel routes. This is why single-deletion FBA cannot see redundancy loss, and why isozyme GPR errors (Bernstein et al., 2023) dominate FBA false negatives.

### C.3 Producibility of a biomass constituent

For metabolite  $i$ , add a demand column  $d_i = -e_i$  to  $S$  and solve

$$\pi_i(F_g) = \max\{u : Sv + d_i u = 0, v \text{ in bounds}, v_j = 0 \text{ for } j \in R(g), u \geq 0\} \quad (C3)$$

Metabolite  $i$  is producible in the knockout iff  $\pi_i > \epsilon$  (we use  $\epsilon = 1e-6$ ). The unproducible set is  $U_g = \{i \text{ in } B : \pi_i(F_g) \leq \epsilon\}$ , where  $B$  is the set of consumed biomass metabolites.

### C.4 The biomass rescue test and its soundness

Write the biomass column as  $b = -\sum_{i \text{ in } B} \beta_i e_i + p$  ( $p$  collects products such as ADP, Pi, H+). The rescued column strips exactly  $U_g$ :

$$b^{(g)} = b + \sum_{i \text{ in } U_g} \beta_i e_i \quad (C4)$$

$$\text{verdict}(g) = \text{biomass\_forced} \text{ iff } z^*(b^{(g)}) \geq \tau z^*, \text{ else network\_forced} \quad (C5)$$

Proposition 1 (soundness of `biomass_forced`). If `verdict(g) = biomass_forced`, then the lethality of  $g$  in the original model is entirely attributable to the requirement for  $U_g$ : the knockout network can carry biomass flux at  $\geq \tau$  of wild type for every other constituent simultaneously. Proof: the rescued LP is

feasible with value  $\geq \tau z^*$ , and its optimal  $v$  is a flux distribution in  $F_g$  that produces each  $i$  in  $B \setminus U_g$  at rate  $\beta_i z^* (b^{(g)})$ . Since the original knockout optimum is zero while the only difference between the two LPs is the  $U_g$  coefficients,  $U_g$  is a sufficient cause of lethality.

Proposition 2 (network\_forced is conservative). If  $\text{verdict}(g) = \text{network\_forced}$ , then even an objective that demands none of the unproducible compounds cannot be met; the lesion is in shared precursor or energy supply. Proof: by (C1) applied to the rescued objective. Note that network\_forced does not claim the gene is truly essential in vivo - only that no biomass relabelling rescues it.

Proposition 3 (minimality).  $U_g$  is minimal among sets defined by single-compound producibility: removing any  $i$  in  $U_g$  from the strip set leaves a compound with  $\pi_i = 0$  in the objective, and a biomass reaction consuming an unproducible compound at positive coefficient forces  $z^* = 0$ . Hence every element of  $U_g$  is necessary for rescue.

Falsifiable prediction. If biomass\_forced labels reflect model artifacts rather than biology, genes so labelled should be enriched for experimentally non-essential genes relative to network\_forced genes. This is tested with the one-sided Fisher exact test (C18) in Section 4.4.

### C.5 AUROC as a Mann-Whitney statistic

For scores  $s$ , positives  $P$  (essential,  $n_1$ ) and negatives  $N$  ( $n_0$ ):

$$AUROC = (1/(n_1 n_0)) \sum_{i \in P} \sum_{j \in N} [1(s_i > s_j) + \frac{1}{2} 1(s_i = s_j)] \quad (C6)$$

Derivation: the ROC curve is  $(FPR(t), TPR(t))$  as  $t$  sweeps; integrating  $TPR \, dFPR$  over the step function gives, for each negative  $j$ , the fraction of positives scoring above it, which summed over  $j$  and normalized is (C6). Ties receive half credit because a tied pair lies on a diagonal ROC segment. Consequence for FBA: binary FBA scores have massive ties, so AUROC collapses to  $(TPR + TNR)/2$  at the single threshold, which explains why pure FBA AUROCs cluster near 0.63-0.67.

$$AUROC_{binary} = \frac{1}{2}(TPR + 1 - FPR) \quad (C7)$$

### C.6 Average precision

$$AP = \sum_k (R_k - R_{k-1}) P_k \quad (C8)$$

with  $P_k, R_k$  precision and recall at the  $k$ -th threshold. The baseline for a random ranker equals the prevalence  $\pi$  (0.172 here), so AUPRC must always be read against  $\pi$ . The lift  $AP/\pi$  is reported alongside.

### C.7 Matthews correlation coefficient

$$MCC = (TP \, TN - FP \, FN) / \sqrt{((TP+FP)(TP+FN)(TN+FP)(TN+FN))} \quad (C9)$$

MCC equals the Pearson correlation of the two binary indicator vectors (proof: expand the covariance of Bernoulli variables), so it is symmetric in classes and insensitive to prevalence in the sense that a constant predictor scores 0.

### C.8 Percentile bootstrap confidence intervals

For statistic T and B resamples of genes with replacement,  $T_b^* = T(X_b^*)$ . The 95% interval is

$$CI_{95} = [ Q_{0.025}(T^*), Q_{0.975}(T^*) ] \quad (C10)$$

Validity rests on the bootstrap distribution of  $T^* - T$  approximating that of  $T - \theta$ ; for U-statistics such as AUROC this holds asymptotically (Hall, 1992). Resamples with a single class are discarded (the code reports `n_boot_used`). Genes are the resampling unit, so the interval reflects gene-set uncertainty, not experimental replicate noise.

### C.9 Paired comparison of two classifiers: McNemar

On paired hard calls, let b = number of genes model A gets right and B wrong, c the reverse. Under  $H_0$  (equal error rates) each discordant gene is a fair coin,  $b \sim \text{Bin}(b+c, 1/2)$ :

$$p = 2 \sum_{k=0}^{\min(b,c)} C(b+c, k) 2^{-(b+c)} \quad (C11)$$

$$\chi^2 = (|b - c| - 1)^2 / (b + c) \quad (C12)$$

(C12) is the continuity-corrected normal approximation, derived from the variance  $(b+c)/4$  of  $\text{Bin}(b+c, 1/2)$ . With  $b = 27$ ,  $c = 50$  the exact test gives  $p = 0.012$  against the ensemble at the 0.5 cut, which is why the paper reports the ranking gain and the hard-call loss together.

### C.10 DeLong-free AUROC difference by paired bootstrap

$$\Delta = AUROC_A - AUROC_B, \quad CI \text{ from paired resamples (same gene indices for A and B)} \quad (C13)$$

Pairing preserves the correlation between models scored on identical genes and is strictly tighter than comparing two independent intervals.

### C.11 Stacked generalization without leakage

Base scores are produced out-of-fold: for fold k with held-out set  $I_k$ , model  $f^{(-k)}$  is trained on the complement and scores  $I_k$ . The meta-learner is

$$h(x) = \sigma( w_0 + \sum_m w_m f_m^{(-k(x))}(x) ) \quad (C14)$$

Because each base prediction for gene x was made without seeing x, the meta-features are distributed as test-time features, and the meta-learner's own OOF evaluation is unbiased up to fold-to-fold variance (Wolpert, 1992). Non-learned FBA scores need no folding.

### C.12 Logistic loss and its gradient

$$L(w) = -\sum_i [ y_i \log p_i + (1-y_i) \log(1-p_i) ] + \lambda ||w||^2/2, \quad \nabla L = X^T(p - y) + \lambda w \quad (C15)$$

Derivation:  $d/dz \log \sigma(z) = 1 - \sigma(z)$ . The Hessian  $X^T \text{diag}(p(1-p))X + \lambda I$  is positive definite, so the meta-learner has a unique optimum.

### C.13 Class-weighted loss for the CNN and GCN

$$L_w = -\sum_i [ \omega_1 y_i \log p_i + \omega_0 (1-y_i) \log(1-p_i) ], \quad \omega_c = n/(2 n_c) \quad (C16)$$

With  $\omega_c = n/(2n_c)$  each class contributes equal total weight, which is the minimizer of expected balanced error for a calibrated model.

#### C.14 Receptive field of the sequence CNN

$$RF_L = 1 + \sum_{l=1}^L (k_l - 1) \prod_{j<l} s_j \quad (C17)$$

For kernel sizes  $k_l$  and strides  $s_j$  (pooling counted as stride). The receptive field bounds the longest motif the network can represent before global pooling; global max pooling then makes the gene score invariant to motif position.

#### C.15 Fisher exact test for provenance enrichment

Let the 2 x 2 table be [[a, b], [c, d]] with rows biomass\_forced / network\_forced and columns non-essential / essential in vivo. Conditioning on margins, a follows the hypergeometric law

$$P(A = a) = C(a+b, a) C(c+d, c) / C(n, a+c) \quad (C18)$$

$$P_{one-sided} = \sum_{k \geq a} P(A = k), \quad OR = ad / bc \quad (C19)$$

The one-sided alternative encodes the prediction of C.4: biomass\_forced calls are more often wrong (non-essential in vivo) than network\_forced calls.

#### C.16 Graph convolution as normalized propagation

$$H^{(l+1)} = \sigma(D^{-1/2} A D^{-1/2} H^{(l)} W^{(l)}), \quad A \blacksquare = A + I \quad (C20)$$

The symmetric normalization has spectrum in (-1, 1], so repeated propagation is contractive and does not explode; two layers aggregate the 2-hop gene neighbourhood via shared metabolites.

#### C.17 Dynamic FBA

$$dX/dt = \mu(t) X, \quad dS_k/dt = v_k(t) X, \quad \mu(t) = \max c^T v \text{ s.t. } Sv = 0, v_k \geq -V_{max,k} S_k / (K_{m,k} + S_k) \quad (C21)$$

$$X_{t+\Delta t} = X_t e^{\mu \Delta t}, \quad S_{t+\Delta t} = S_t + v (X_t / \mu) (e^{\mu \Delta t} - 1) \quad (C22)$$

(C22) is the exact solution of (C21) with fluxes frozen over the step (the static-optimization approach of Mahadevan et al., 2002); it avoids the negative-concentration artifacts of forward Euler. The Boolean catabolite-repression overlay sets  $V_{max,lac} = 0$  while  $S_{glc} > \theta$ , producing diauxie.

#### C.18 Threshold sensitivity and bimodality

$$E(\tau) = 1(r_g < \tau), \quad \partial |E(\tau)| / \partial \tau = \sum_g \delta(r_g - \tau) \quad (C23)$$

Because FBA growth fractions are almost all exactly 0 or 1, the density of  $r_g$  in (0.001, 0.5) is nearly empty and  $|E(\tau)|$  is flat across two orders of magnitude of  $\tau$  - the robustness result of Section 3.

#### C.19 Lift and expected calibration

$$lift = AP / \pi, \quad ECE = \sum_b (|B_b|/n) |acc(B_b) - conf(B_b)| \quad (C24)$$

Lift normalizes AUPRC for prevalence, which differs between Gerdes (17.2%) and the PEC subset, making cross-yardstick comparisons meaningful.

## Appendix D. Extended evaluation: distributions, calibration and early precision

AUROC summarizes the whole ranking, but a biologist using a virtual cell reads it from the top: which genes should be tested first? This section reports the views of the same out-of-fold scores that answer that question. Every panel is generated by scripts/make\_figures\_ext.py directly from results/aligned\_predictions\_all\_models.csv and results/ensemble\_oof.csv; no number is transcribed by hand.



Figure 6. Out-of-fold score distributions by Gerdes class for the GCN, sequence CNN, k-mer baseline and stacked ensemble. Only the ensemble develops a distinct high-score essential mode (scores > 0.8).

Figure 6 shows why the ensemble beats every member on ranking. Each learned base model shifts the essential class to the right but leaves the two classes overlapping across the full score range. The ensemble, which also receives the binary FBA calls, splits the essential genes into two groups: a sharp mode near 0.85-0.95 made of genes that FBA calls essential AND the learned models score high, and a broad low mode of essential genes that FBA misses. The second group is the ceiling of this architecture - no amount of stacking recovers genes that none of the inputs can see.

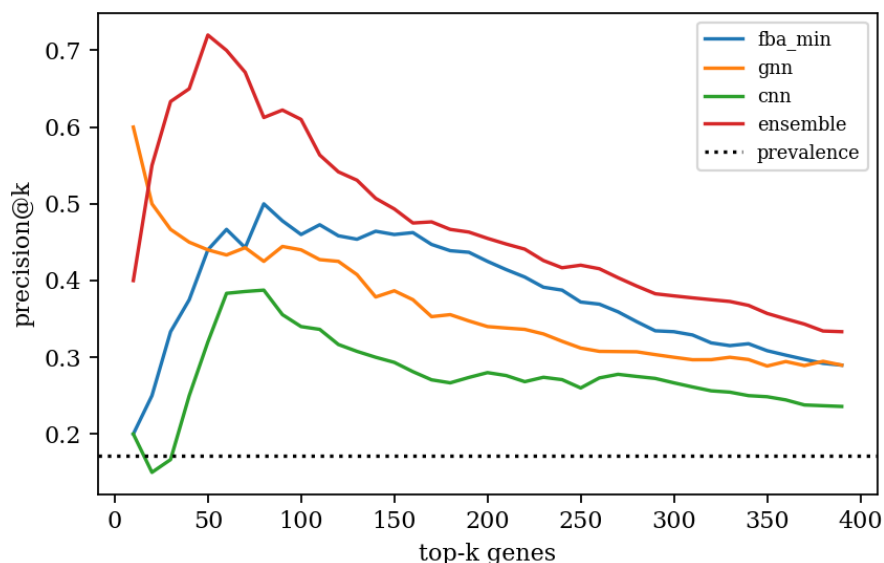


Figure 10. Precision among the top-k ranked genes. Dotted line: prevalence (0.172).

| model    | P@25 | P@50 | P@100 | P@200 |
|----------|------|------|-------|-------|
| fba_min  | 0.32 | 0.44 | 0.46  | 0.42  |
| gnn      | 0.52 | 0.44 | 0.44  | 0.34  |
| cnn      | 0.12 | 0.32 | 0.34  | 0.28  |
| kmer     | 0.44 | 0.34 | 0.38  | 0.33  |
| ensemble | 0.60 | 0.72 | 0.61  | 0.46  |

Table 8. Precision at  $k$  (fraction of the top- $k$  genes that are essential in Gerdes 2003).

At  $k = 50$  the ensemble reaches precision 0.72, 4.2x the prevalence and well above the best single model (FBA minimal 0.44, GCN 0.44). The advantage shrinks with  $k$ : by  $k = 200$  the ensemble (0.46) is only marginally ahead of FBA (0.42). The practical reading is that the ensemble is a good triage tool for the first 50-100 experiments and not a replacement for FBA deeper in the list. The early dip in the ensemble curve at  $k = 10$  is real and explained in Section 4: the very top of the ranking is occupied by molybdopterin genes (*moaD*, *moaE*, *moaA*, *moaC*) that every component scores as essential and that Gerdes finds dispensable - the biomass-objective artifact analysed in Section 4.3.



Figure 7. Reliability diagram (quantile bins; scores min-max scaled per model).

Figure 7 shows that none of the scores is calibrated as a probability: all curves lie below the diagonal, meaning that scaled scores overstate essentiality. This is expected for class-weighted training (Appendix C, eq. C16), which trades calibration for balanced error. The ensemble curve is the closest to monotone and reaches an observed essential fraction of about 0.55 in its top decile. For downstream use as probabilities, a Platt or isotonic recalibration on a held-out split is required; we do not claim calibrated probabilities anywhere in this paper.



Figure 8. Per-fold AUROC for the GCN and the sequence CNN (3-fold stratified CV).

Fold-to-fold spread (Figure 8) is about 0.05 AUROC for the GCN and 0.02 for the CNN. The GCN spread is comparable to the gap between the GCN and FBA, so the ordering GCN > FBA-rich in Table 1 should be read as a tie within fold noise. The ensemble gain (+0.057 over FBA minimal) is larger than both spreads and survives the paired bootstrap of Appendix C.10.



Figure 9. F1 and number of predicted essential genes versus the growth-fraction threshold.

Figure 9 is the graphical form of the bimodality result (eq. C23): both curves are flat, since exactly 183 genes are called essential at 0.5%, 1% and 5% of wild-type growth.

Table 9 lists the 40 highest-ranked genes so that any reader can check the ranking against EcoCyc or their own data. False positives cluster in cofactor biosynthesis (molybdopterin, folate), which motivated the provenance audit of Section 4.4.

| rank | b-number | gene | ensemble | FBA min | GCN  | Gerdes   |
|------|----------|------|----------|---------|------|----------|
| 1    | b0784    | moaD | 0.963    | E       | 0.81 | non-ess. |
| 2    | b0785    | moaE | 0.958    | E       | 0.76 | non-ess. |

| rank | b-number | gene  | ensemble | FBA min | GCN  | Gerdess   |
|------|----------|-------|----------|---------|------|-----------|
| 3    | b2746    | ispF  | 0.952    | E       | 0.72 | essential |
| 4    | b0142    | folK  | 0.948    | E       | 0.81 | essential |
| 5    | b0009    | mog   | 0.944    | E       | 0.71 | essential |
| 6    | b3857    | mobA  | 0.944    | E       | 0.82 | non-ess.  |
| 7    | b3018    | plsC  | 0.943    | E       | 0.80 | essential |
| 8    | b0783    | moaC  | 0.939    | E       | 0.75 | non-ess.  |
| 9    | b3058    | folB  | 0.938    | E       | 0.72 | non-ess.  |
| 10   | b3177    | folP  | 0.934    | E       | 0.79 | non-ess.  |
| 11   | b3634    | coaD  | 0.930    | E       | 0.67 | essential |
| 12   | b2747    | ispD  | 0.928    | E       | 0.76 | essential |
| 13   | b3807    | cyaY  | 0.927    | E       | 0.71 | non-ess.  |
| 14   | b0415    | ribH  | 0.925    | E       | 0.71 | essential |
| 15   | b0029    | ispH* | 0.925    | E       | 0.69 | essential |
| 16   | b0159    | mtn   | 0.922    | E       | 0.48 | essential |
| 17   | b0781    | moaA  | 0.921    | E       | 0.70 | non-ess.  |
| 18   | b3041    | ribB  | 0.920    | E       | 0.49 | essential |
| 19   | b0369    | hemB  | 0.919    | E       | 0.58 | essential |
| 20   | b1096    | pabC  | 0.919    | E       | 0.59 | non-ess.  |
| 21   | b1208    | ispE  | 0.918    | E       | 0.71 | essential |
| 22   | b0826    | moeB  | 0.917    | E       | 0.67 | non-ess.  |
| 23   | b1277    | ribA  | 0.917    | E       | 0.44 | essential |
| 24   | b0421    | ispA  | 0.915    | E       | 0.69 | essential |
| 25   | b0174    | uppS  | 0.913    | E       | 0.54 | essential |
| 26   | b3648    | gmk   | 0.909    | E       | 0.61 | essential |
| 27   | b3639    | dfp   | 0.908    | E       | 0.60 | essential |
| 28   | b3040    | zupT  | 0.907    | E       | 0.43 | non-ess.  |
| 29   | b1098    | tmk   | 0.905    | E       | 0.65 | essential |
| 30   | b0103    | coaE  | 0.904    | E       | 0.70 | essential |
| 31   | b1662    | ribE  | 0.903    | E       | 0.66 | essential |
| 32   | b0154    | hemL  | 0.902    | E       | 0.54 | essential |
| 33   | b2515    | ispG* | 0.901    | E       | 0.73 | essential |
| 34   | b1415    | aldA  | 0.894    | E       | 0.48 | non-ess.  |
| 35   | b0417    | thiL  | 0.894    | E       | 0.62 | essential |
| 36   | b4039    | ubiC  | 0.893    | E       | 0.48 | non-ess.  |
| 37   | b0475    | hemH  | 0.893    | E       | 0.83 | essential |
| 38   | b0087    | mraY  | 0.890    | E       | 0.56 | non-ess.  |
| 39   | b2530    | iscS  | 0.890    | E       | 0.53 | essential |
| 40   | b2585    | pssA  | 0.889    | E       | 0.75 | essential |

Table 9. Top 40 genes by out-of-fold ensemble score with the Gerdess 2003 call.

## Appendix E. Essentiality-provenance catalog for iJO1366

### E.1 Question and design

Section 4.3 showed one case - the molybdopterin cluster - where an FBA essentiality call is produced by the biomass objective rather than by the network. This appendix asks the general question: for every one of the 208 genes that iJO1366 calls essential on glucose minimal medium, is the call network\_forced or biomass\_forced? The rescue test (Appendix C.4, eqs. C3-C5) answers it per gene with one re-runnable LP experiment. The tool is shipped as `python -m vcell audit-biomass MODEL` and runs on any COBRA model file.

Prior work. The phenomenon of cofactor-driven false essentiality is known. Bernstein et al. (2023, Mol Syst Biol) found 21 vitamin/cofactor biosynthesis genes in iML1515 (biotin, R-pantothenate, thiamin, tetrahydrofolate, NAD+) that the model calls essential and RB-TnSeq finds fit, and fixed them by manually supplying those cofactors. MEMOTE tests whether biomass precursors are producible in the wild-type model but does not attribute any knockout's lethality to specific compounds. Our contribution is narrower and procedural: an automated, per-gene, falsifiable attribution. We found no existing tool that does this; that is a statement about our search, not a proof of absence.

### E.2 Verdict census

| verdict        | genes |
|----------------|-------|
| biomass_forced | 208   |

Table 10. Rescue-test verdicts.

Negative result 1: the binary verdict does not discriminate. All 208 iJO1366 essential genes are biomass\_forced. Every lethal knockout on glucose minimal medium makes at least one biomass constituent unproducible, and removing those constituents always restores some growth. Proposition 1 is true but uninformative here: the set  $U_g$  ranges from 1 to 50 compounds (median 2), and when  $U_g$  is large the rescued objective is a different, much easier objective. The rescue test therefore needs a second axis before it says anything about artifacts.

### E.3 A methods catch: raw biomass\_forced over-calls artifacts

A raw biomass\_forced label means only that stripping the unproducible compounds rescues growth. For a gene such as *fabD* that stops lipid synthesis this is trivially true and says nothing about artifacts: lipids are really required. The artifact hypothesis only makes sense for compounds a mutant might plausibly obtain without making them (trace cofactors and vitamins by carry-over or cross-feeding, the class named by Bernstein et al.). We therefore split biomass constituents into cofactor/vitamin, inorganic and bulk classes (`vcell/compound_classes.py`) and label a gene *cofactor\_only* when every unproducible compound is a cofactor or vitamin.

| stratum                      | genes | Gerdes essential | fraction |
|------------------------------|-------|------------------|----------|
| biomass_forced:bulk_involved | 104   | 39               | 0.38     |
| biomass_forced:cofactor_only | 79    | 42               | 0.53     |

Table 11. Gerdes 2003 essentiality by provenance stratum.

#### E.4 The falsifiable test

Prediction (Appendix C.4): cofactor\_only biomass\_forced genes are enriched for experimentally non-essential genes relative to all other FBA-essential genes. One-sided Fisher exact test (eqs. C18-C19) on [[non-ess, ess], [non-ess, ess]] = [[37, 42], [65, 39]]: odds ratio 0.53,  $p = 0.988$ . For the raw (unstratified) biomass\_forced label the table is [[102, 81], [0, 0]] (degenerate: the network\_forced row is empty, so no test is possible): odds ratio nan,  $p = 1$ .

Negative result 2: the cofactor-artifact hypothesis fails on this yardstick. Cofactor-only genes are more often essential in Gerdes 2003 than bulk-precursor genes, the opposite of the prediction, and the one-sided test is nowhere near significant. The likely reason is a condition mismatch. Gerdes screened on rich LB medium, where amino acids and nucleotide precursors are supplied, so bulk-precursor biosynthesis genes are dispensable, while most cofactors are not supplied at useful levels, so cofactor genes stay essential. The model was simulated on glucose minimal medium. A condition-matched test (RB-TnSeq on defined carbon sources, Appendix H) is the right place to test the hypothesis. We keep the failed test in the paper because it shows that the yardstick has to match the condition before essentiality errors can be attributed.

#### E.5 Corrected benchmark

| metric    | FBA iJO1366 (original) | after cofactor_only correction |
|-----------|------------------------|--------------------------------|
| auroc     | 0.666                  | 0.584                          |
| auprc     | 0.316                  | 0.247                          |
| precision | 0.437                  | 0.369                          |
| recall    | 0.386                  | 0.191                          |
| f1        | 0.410                  | 0.252                          |
| mcc       | 0.297                  | 0.163                          |

Table 12. iJO1366 minimal-medium FBA vs Gerdes 2003 before and after flipping cofactor\_only biomass\_forced genes to non-essential.

Corrected AUROC 95% CI: [0.544, 0.626]. The correction uses only the audit, never the Gerdes labels, and it makes the model worse on this yardstick, consistent with negative result 2. It is not adopted.

| biomass compound | genes whose knockout makes it unproducible |
|------------------|--------------------------------------------|
| btn_c            | 37                                         |
| bmocogdp_c       | 28                                         |
| thmpp_c          | 28                                         |
| coa_c            | 27                                         |
| 10fthf_c         | 26                                         |
| mlthf_c          | 26                                         |
| thf_c            | 26                                         |
| amet_c           | 23                                         |

| biomass compound | genes whose knockout makes it unproducible |
|------------------|--------------------------------------------|
| murein5px4p_p    | 22                                         |
| fad_c            | 19                                         |
| nadp_c           | 17                                         |
| kdo2lipid4_e     | 17                                         |
| ribflv_c         | 17                                         |
| 2ohph_c          | 16                                         |
| nad_c            | 16                                         |
| dctp_c           | 12                                         |
| gtp_c            | 12                                         |
| met__L_c         | 12                                         |
| atp_c            | 11                                         |
| datp_c           | 11                                         |
| his__L_c         | 11                                         |
| pheme_c          | 10                                         |
| sheme_c          | 10                                         |
| udcpdp_c         | 9                                          |
| trp__L_c         | 9                                          |
| lys__L_c         | 8                                          |
| arg__L_c         | 8                                          |
| phe__L_c         | 8                                          |
| 2fe2s_c          | 8                                          |
| 4fe4s_c          | 8                                          |
| leu__L_c         | 7                                          |
| tyr__L_c         | 7                                          |
| dttp_c           | 7                                          |
| cys__L_c         | 7                                          |
| ile__L_c         | 6                                          |
| ctp_c            | 6                                          |
| dctp_c           | 6                                          |
| pydx5p_c         | 5                                          |
| pe160_p          | 5                                          |
| pe161_p          | 5                                          |
| utp_c            | 5                                          |
| pe160_c          | 4                                          |
| pe161_c          | 4                                          |
| thr__L_c         | 3                                          |
| pro__L_c         | 3                                          |
| gln__L_c         | 2                                          |
| glu__L_c         | 2                                          |

| biomass compound | genes whose knockout makes it unproducible |
|------------------|--------------------------------------------|
| val__L_c         | 2                                          |
| asn__L_c         | 1                                          |
| asp__L_c         | 1                                          |
| cobalt2_c        | 1                                          |
| cu2_c            | 1                                          |
| ca2_c            | 1                                          |

Table 13. Compound census over biomass\_forced genes.

## E.6 Full catalog

Every audited gene with its verdict, the unproducible biomass compounds (truncated to 60 characters) and its Gerdes 2003 call (E essential, N non-essential, n/a not in the aligned set). \* marks network\_forced\_no\_biomass\_link: no individual biomass constituent becomes unproducible, yet growth stops, which points to a coupling or energy constraint.

| b-number | gene  | verdict        | unproducible compounds   | Gerdes |
|----------|-------|----------------|--------------------------|--------|
| b0003    |       | biomass_forced | ile__L_c, thr__L_c       | n/a    |
| b0004    | thrC  | biomass_forced | ile__L_c, thr__L_c       | N      |
| b0009    | mog   | biomass_forced | bmocogdp_c               | E      |
| b0025    | ribF  | biomass_forced | fad_c                    | E      |
| b0029    | ispH* | biomass_forced | 2ohph_c, ucdpdp_c        | E      |
| b0031    | dapB  | biomass_forced | lys__L_c, murein5px4p_p  | E      |
| b0052    | pdxA  | biomass_forced | pydx5p_c                 | N      |
| b0071    | leuD  | biomass_forced | leu__L_c                 | N      |
| b0072    | leuC  | biomass_forced | leu__L_c                 | N      |
| b0073    | leuB  | biomass_forced | leu__L_c                 | N      |
| b0074    | leuA  | biomass_forced | leu__L_c                 | N      |
| b0085    | murE  | biomass_forced | murein5px4p_p            | E      |
| b0086    | murF  | biomass_forced | murein5px4p_p            | N      |
| b0087    | mraY  | biomass_forced | murein5px4p_p            | N      |
| b0088    | murD  | biomass_forced | murein5px4p_p            | E      |
| b0090    | murG  | biomass_forced | murein5px4p_p            | E      |
| b0091    | murC  | biomass_forced | murein5px4p_p            | E      |
| b0096    |       | biomass_forced | kdo2lipid4_e             | n/a    |
| b0103    | coaE  | biomass_forced | coa_c                    | E      |
| b0109    | nadC  | biomass_forced | nad_c, nadp_c            | N      |
| b0131    | panD  | biomass_forced | coa_c                    | N      |
| b0133    | panC  | biomass_forced | coa_c                    | N      |
| b0134    | panB  | biomass_forced | coa_c                    | N      |
| b0142    | folK  | biomass_forced | 10fthf_c, mlthf_c, thf_c | E      |
| b0154    | hemL  | biomass_forced | pheme_c, sheme_c         | E      |

| b-number | gene | verdict        | unproducible compounds                                       | Gerdas |
|----------|------|----------------|--------------------------------------------------------------|--------|
| b0159    | mtn  | biomass_forced | btn_c, sheme_c, thmpp_c                                      | E      |
| b0166    | dapD | biomass_forced | lys__L_c, murein5px4p_p                                      | E      |
| b0173    | dxr  | biomass_forced | 2ohph_c, ucdpdp_c                                            | E      |
| b0174    | uppS | biomass_forced | udcpdp_c                                                     | E      |
| b0175    | cdsA | biomass_forced | pe160_c, pe160_p, pe161_c, pe161_p                           | E      |
| b0179    |      | biomass_forced | kdo2lipid4_e                                                 | n/a    |
| b0180    | fabZ | biomass_forced | btn_c                                                        | E      |
| b0181    | lpxA | biomass_forced | kdo2lipid4_e                                                 | E      |
| b0182    | lpxB | biomass_forced | kdo2lipid4_e                                                 | E      |
| b0185    | accA | biomass_forced | btn_c, kdo2lipid4_e                                          | E      |
| b0369    | hemB | biomass_forced | pheme_c, sheme_c                                             | E      |
| b0386    | proC | biomass_forced | pro__L_c                                                     | E      |
| b0414    | ribD | biomass_forced | fad_c, ribflv_c                                              | E      |
| b0415    | ribH | biomass_forced | fad_c, ribflv_c                                              | E      |
| b0417    | thiL | biomass_forced | thmpp_c                                                      | E      |
| b0420    | dxs  | biomass_forced | 2ohph_c, pydx5p_c, thmpp_c, ucdpdp_c                         | E      |
| b0421    | ispA | biomass_forced | 2ohph_c, ucdpdp_c                                            | E      |
| b0423    | thiI | biomass_forced | thmpp_c                                                      | N      |
| b0475    | hemH | biomass_forced | pheme_c                                                      | E      |
| b0522    | purK | biomass_forced | 10fthf_c, amet_c, atp_c, bmocogdp_c, btn_c, coa_c, datp_c, d | N      |
| b0523    | purE | biomass_forced | 10fthf_c, amet_c, atp_c, bmocogdp_c, btn_c, coa_c, datp_c, d | N      |
| b0524    | lpxH | biomass_forced | kdo2lipid4_e                                                 | E      |
| b0639    | nadD | biomass_forced | nad_c, nadp_c                                                | N      |
| b0720    | gltA | biomass_forced | 10fthf_c, arg__L_c, gln__L_c, glu__L_c, mlthf_c, murein5px4p | N      |
| b0750    | nadA | biomass_forced | nad_c, nadp_c                                                | N      |
| b0774    | bioA | biomass_forced | btn_c                                                        | N      |
| b0775    | bioB | biomass_forced | btn_c                                                        | N      |
| b0776    | bioF | biomass_forced | btn_c                                                        | N      |
| b0777    | bioC | biomass_forced | btn_c                                                        | N      |
| b0778    | bioD | biomass_forced | btn_c                                                        | N      |
| b0781    | moaA | biomass_forced | bmocogdp_c                                                   | N      |
| b0783    | moaC | biomass_forced | bmocogdp_c                                                   | N      |
| b0784    | moaD | biomass_forced | bmocogdp_c                                                   | N      |
| b0785    | moaE | biomass_forced | bmocogdp_c                                                   | N      |
| b0826    | moeB | biomass_forced | bmocogdp_c                                                   | N      |
| b0827    | moeA | biomass_forced | bmocogdp_c                                                   | N      |
| b0907    | serC | biomass_forced | pydx5p_c                                                     | N      |
| b0908    | aroA | biomass_forced | 10fthf_c, 2ohph_c, mlthf_c, phe__L_c, thf_c, thmpp_c, trp__L | N      |
| b0914    | msbA | biomass_forced | kdo2lipid4_e, pe160_p, pe161_p                               | E      |

| <b>b-number</b> | <b>gene</b> | <b>verdict</b> | <b>unproducible compounds</b>                                | <b>Gerdas</b> |
|-----------------|-------------|----------------|--------------------------------------------------------------|---------------|
| b0915           | lpxK        | biomass_forced | kdo2lipid4_e                                                 | E             |
| b0918           | kdsB        | biomass_forced | kdo2lipid4_e                                                 | E             |
| b0928           | aspC        | biomass_forced | 10fthf_c, amet_c, arg__L_c, asn__L_c, asp__L_c, atp_c, bmoco | N             |
| b1062           | pyrC        | biomass_forced | ctp_c, dctp_c, dttp_c, utp_c                                 | N             |
| b1069           | mviN        | biomass_forced | murein5px4p_p                                                | E             |
| b1091           | fabH        | biomass_forced | btn_c                                                        | E             |
| b1092           | fabD        | biomass_forced | btn_c, kdo2lipid4_e                                          | E             |
| b1093           | fabG        | biomass_forced | btn_c, kdo2lipid4_e                                          | E             |
| b1094           |             | biomass_forced | btn_c, kdo2lipid4_e, pe160_c, pe160_p, pe161_c, pe161_p      | n/a           |
| b1096           | pabC        | biomass_forced | 10fthf_c, mlthf_c, thf_c                                     | N             |
| b1098           | tmk         | biomass_forced | dttp_c                                                       | E             |
| b1131           | purB        | biomass_forced | 10fthf_c, amet_c, atp_c, bmocogdp_c, btn_c, coa_c, datp_c, d | N             |
| b1136           | icd         | biomass_forced | 10fthf_c, arg__L_c, gln__L_c, glu__L_c, mlthf_c, murein5px4p | N             |
| b1208           | ispE        | biomass_forced | 2ohph_c, ucdpdp_c                                            | E             |
| b1210           |             | biomass_forced | pheme_c, sheme_c                                             | n/a           |
| b1215           |             | biomass_forced | kdo2lipid4_e                                                 | n/a           |
| b1260           | trpA        | biomass_forced | trp__L_c                                                     | N             |
| b1261           | trpB        | biomass_forced | trp__L_c                                                     | N             |
| b1262           | trpC        | biomass_forced | trp__L_c                                                     | N             |
| b1263           | trpD        | biomass_forced | trp__L_c                                                     | N             |
| b1264           | trpE        | biomass_forced | trp__L_c                                                     | N             |
| b1277           | ribA        | biomass_forced | fad_c, ribflv_c                                              | E             |
| b1281           | pyrF        | biomass_forced | ctp_c, dctp_c, dttp_c, utp_c                                 | E             |
| b1288           | fabI        | biomass_forced | btn_c                                                        | E             |
| b1415           | aldA        | biomass_forced | 10fthf_c, mlthf_c, thf_c                                     | N             |
| b1662           | ribE        | biomass_forced | fad_c, ribflv_c                                              | E             |
| b1693           | aroD        | biomass_forced | 10fthf_c, 2ohph_c, mlthf_c, phe__L_c, thf_c, thmpp_c, trp__L | N             |
| b1740           | nadE        | biomass_forced | nad_c, nadp_c                                                | E             |
| b1812           | pabB        | biomass_forced | 10fthf_c, mlthf_c, thf_c                                     | N             |
| b2019           |             | biomass_forced | his__L_c                                                     | n/a           |
| b2020           | hisD        | biomass_forced | his__L_c                                                     | N             |
| b2021           | hisC        | biomass_forced | his__L_c                                                     | N             |
| b2022           | hisB        | biomass_forced | his__L_c                                                     | N             |
| b2023           | hisH        | biomass_forced | his__L_c                                                     | N             |
| b2024           | hisA        | biomass_forced | his__L_c                                                     | N             |
| b2025           | hisF        | biomass_forced | his__L_c                                                     | N             |
| b2026           | hisI        | biomass_forced | his__L_c                                                     | N             |
| b2103           | thiD        | biomass_forced | thmpp_c                                                      | N             |
| b2153           | folE        | biomass_forced | 10fthf_c, mlthf_c, thf_c                                     | E             |

| b-number | gene  | verdict        | unproducible compounds                                        | Gerdes |
|----------|-------|----------------|---------------------------------------------------------------|--------|
| b2311    | ubiX  | biomass_forced | 2ohph_c                                                       | E      |
| b2312    | purF  | biomass_forced | 10fthf_c, amet_c, atp_c, bmocogdp_c, btn_c, coa_c, datp_c, d  | N      |
| b2315    |       | biomass_forced | 10fthf_c, mlthf_c, thf_c                                      | n/a    |
| b2316    |       | biomass_forced | btn_c, kdo2lipid4_e                                           | n/a    |
| b2320    | pdxB  | biomass_forced | pydx5p_c                                                      | N      |
| b2323    | fabB  | biomass_forced | btn_c, kdo2lipid4_e                                           | E      |
| b2329    | aroC  | biomass_forced | 10fthf_c, 2ohph_c, mlthf_c, phe__L_c, thf_c, thmpp_c, trp__L  | N      |
| b2400    | gltX  | biomass_forced | pheme_c, sheme_c                                              | E      |
| b2472    | dapE  | biomass_forced | lys__L_c, murein5px4p_p                                       | E      |
| b2476    | purC  | biomass_forced | 10fthf_c, amet_c, atp_c, bmocogdp_c, btn_c, coa_c, datp_c, d  | N      |
| b2478    | dapA  | biomass_forced | lys__L_c, murein5px4p_p                                       | E      |
| b2499    | purM  | biomass_forced | 10fthf_c, amet_c, atp_c, bmocogdp_c, btn_c, coa_c, datp_c, d  | N      |
| b2507    | guaA  | biomass_forced | 10fthf_c, bmocogdp_c, dgtp_c, fad_c, gtp_c, mlthf_c, ribflv__ | N      |
| b2515    | ispG* | biomass_forced | 2ohph_c, ucdpdp_c                                             | E      |
| b2530    | iscS  | biomass_forced | bmocogdp_c, thmpp_c                                           | E      |
| b2557    | purL  | biomass_forced | 10fthf_c, amet_c, atp_c, bmocogdp_c, btn_c, coa_c, datp_c, d  | N      |
| b2564    | pdxJ  | biomass_forced | pydx5p_c                                                      | E      |
| b2574    | nadB  | biomass_forced | nad_c, nadp_c                                                 | N      |
| b2585    | pssA  | biomass_forced | pe160_c, pe160_p, pe161_c, pe161_p                            | E      |
| b2599    | pheA  | biomass_forced | phe__L_c, thmpp_c, tyr__L_c                                   | N      |
| b2600    | tyrA  | biomass_forced | phe__L_c, thmpp_c, tyr__L_c                                   | N      |
| b2615    | ppnK  | biomass_forced | nadp_c                                                        | E      |
| b2687    |       | biomass_forced | btn_c, sheme_c                                                | n/a    |
| b2746    | ispF  | biomass_forced | 2ohph_c, ucdpdp_c                                             | E      |
| b2747    | ispD  | biomass_forced | 2ohph_c, ucdpdp_c                                             | E      |
| b2750    |       | biomass_forced | 2fe2s_c, 4fe4s_c, amet_c, bmocogdp_c, btn_c, coa_c, cys__L_c  | n/a    |
| b2751    | cysN  | biomass_forced | 2fe2s_c, 4fe4s_c, amet_c, bmocogdp_c, btn_c, coa_c, cys__L_c  | N      |
| b2752    | cysD  | biomass_forced | 2fe2s_c, 4fe4s_c, amet_c, bmocogdp_c, btn_c, coa_c, cys__L_c  | N      |
| b2762    | cysH  | biomass_forced | 2fe2s_c, 4fe4s_c, amet_c, bmocogdp_c, btn_c, coa_c, cys__L_c  | N      |
| b2763    | cysI  | biomass_forced | 2fe2s_c, 4fe4s_c, amet_c, bmocogdp_c, btn_c, coa_c, cys__L_c  | N      |
| b2764    | cysJ  | biomass_forced | 2fe2s_c, 4fe4s_c, amet_c, bmocogdp_c, btn_c, coa_c, cys__L_c  | N      |
| b2780    | pyrG  | biomass_forced | ctp_c, dctp_c                                                 | E      |
| b2818    | argA  | biomass_forced | arg__L_c                                                      | N      |
| b2827    | thyA  | biomass_forced | dttp_c                                                        | E      |
| b2838    | lysA  | biomass_forced | lys__L_c                                                      | N      |
| b2942    | metK  | biomass_forced | amet_c, btn_c, sheme_c, thmpp_c                               | E      |
| b3018    | plsC  | biomass_forced | pe160_c, pe160_p, pe161_c, pe161_p                            | E      |
| b3040    | zupT  | biomass_forced | cobalt2_c, cu2_c                                              | N      |
| b3041    | ribB  | biomass_forced | fad_c, ribflv_c                                               | E      |

| b-number | gene | verdict        | unproducible compounds                                        | Gerdas |
|----------|------|----------------|---------------------------------------------------------------|--------|
| b3058    | folB | biomass_forced | 10fthf_c, mlthf_c, thf_c                                      | N      |
| b3172    | argG | biomass_forced | arg__L_c                                                      | N      |
| b3176    | mrsA | biomass_forced | kdo2lipid4_e, murein5px4p_p                                   | N      |
| b3177    | folP | biomass_forced | 10fthf_c, mlthf_c, thf_c                                      | N      |
| b3187    | ispB | biomass_forced | 2ohph_c                                                       | E      |
| b3189    | murA | biomass_forced | murein5px4p_p                                                 | N      |
| b3196    | yrbG | biomass_forced | ca2_c                                                         | N      |
| b3198    |      | biomass_forced | kdo2lipid4_e                                                  | n/a    |
| b3199    |      | biomass_forced | kdo2lipid4_e                                                  | n/a    |
| b3200    |      | biomass_forced | kdo2lipid4_e                                                  | n/a    |
| b3201    | yhbG | biomass_forced | kdo2lipid4_e                                                  | E      |
| b3255    |      | biomass_forced | btn_c, kdo2lipid4_e                                           | n/a    |
| b3256    |      | biomass_forced | btn_c, kdo2lipid4_e                                           | n/a    |
| b3359    |      | biomass_forced | arg__L_c, lys__L_c, murein5px4p_p                             | n/a    |
| b3360    | pabA | biomass_forced | 10fthf_c, mlthf_c, thf_c                                      | N      |
| b3368    |      | biomass_forced | sheme_c                                                       | n/a    |
| b3389    | aroB | biomass_forced | 10fthf_c, 2ohph_c, mlthf_c, phe__L_c, thf_c, thmpp_c, trp__L  | N      |
| b3412    | bioH | biomass_forced | btn_c                                                         | N      |
| b3433    | asd  | biomass_forced | amet_c, btn_c, ile__L_c, lys__L_c, met__L_c, murein5px4p_p,   | E      |
| b3607    | cysE | biomass_forced | 2fe2s_c, 4fe4s_c, amet_c, bmocogdp_c, btn_c, coa_c, cys__L_c  | E      |
| b3608    | gpsA | biomass_forced | pe160_c, pe160_p, pe161_c, pe161_p                            | N      |
| b3633    | kdtA | biomass_forced | kdo2lipid4_e                                                  | E      |
| b3634    | coaD | biomass_forced | coa_c                                                         | E      |
| b3639    | dfp  | biomass_forced | coa_c                                                         | E      |
| b3642    | pyrE | biomass_forced | ctp_c, dctp_c, dttp_c, utp_c                                  | N      |
| b3648    | gmK  | biomass_forced | 10fthf_c, bmocogdp_c, dgtp_c, fad_c, gtp_c, mlthf_c, ribflv__ | E      |
| b3729    | glmS | biomass_forced | kdo2lipid4_e, murein5px4p_p                                   | E      |
| b3730    | glmU | biomass_forced | kdo2lipid4_e, murein5px4p_p                                   | E      |
| b3770    | ilvE | biomass_forced | ile__L_c, leu__L_c, phe__L_c                                  | N      |
| b3771    | ilvD | biomass_forced | coa_c, ile__L_c, leu__L_c, val__L_c                           | N      |
| b3774    | ilvC | biomass_forced | coa_c, ile__L_c, leu__L_c, val__L_c                           | N      |
| b3804    | hemD | biomass_forced | pheme_c, sheme_c                                              | E      |
| b3805    | hemC | biomass_forced | pheme_c, sheme_c                                              | E      |
| b3807    | cyaY | biomass_forced | 2fe2s_c, 4fe4s_c                                              | N      |
| b3809    | dapF | biomass_forced | lys__L_c, murein5px4p_p                                       | E      |
| b3843    | ubiD | biomass_forced | 2ohph_c                                                       | N      |
| b3850    | hemG | biomass_forced | pheme_c                                                       | E      |
| b3857    | mobA | biomass_forced | bmocogdp_c                                                    | N      |
| b3939    | metB | biomass_forced | amet_c, btn_c, met__L_c                                       | N      |

| b-number | gene | verdict        | unproducible compounds                                       | Gerdas |
|----------|------|----------------|--------------------------------------------------------------|--------|
| b3941    | metF | biomass_forced | amet_c, btn_c, met__L_c, sheme_c                             | N      |
| b3957    |      | biomass_forced | arg__L_c                                                     | n/a    |
| b3958    | argC | biomass_forced | arg__L_c                                                     | E      |
| b3959    | argB | biomass_forced | arg__L_c                                                     | N      |
| b3960    | argH | biomass_forced | arg__L_c                                                     | N      |
| b3967    | murI | biomass_forced | murein5px4p_p                                                | E      |
| b3972    | murB | biomass_forced | murein5px4p_p                                                | N      |
| b3974    | coaA | biomass_forced | coa_c                                                        | E      |
| b3990    |      | biomass_forced | thmpp_c                                                      | n/a    |
| b3991    |      | biomass_forced | thmpp_c                                                      | n/a    |
| b3992    | thiF | biomass_forced | thmpp_c                                                      | N      |
| b3993    | thiE | biomass_forced | thmpp_c                                                      | N      |
| b3994    | thiC | biomass_forced | thmpp_c                                                      | N      |
| b3997    | hemE | biomass_forced | pheme_c                                                      | E      |
| b4005    | purD | biomass_forced | 10fthf_c, amet_c, atp_c, bmocogdp_c, btn_c, coa_c, datp_c, d | N      |
| b4006    | purH | biomass_forced | 10fthf_c, amet_c, atp_c, bmocogdp_c, btn_c, coa_c, datp_c, d | N      |
| b4013    | metA | biomass_forced | amet_c, btn_c, met__L_c                                      | N      |
| b4039    | ubiC | biomass_forced | 2ohph_c                                                      | N      |
| b4040    |      | biomass_forced | 2ohph_c                                                      | n/a    |
| b4160    |      | biomass_forced | pe160_c, pe160_p, pe161_c, pe161_p                           | n/a    |
| b4177    | purA | biomass_forced | amet_c, atp_c, btn_c, coa_c, datp_c, fad_c, his__L_c, nad_c, | N      |
| b4214    | cysQ | biomass_forced | 2fe2s_c, 4fe4s_c, amet_c, bmocogdp_c, btn_c, coa_c, cys__L_c | N      |
| b4245    | pyrB | biomass_forced | ctp_c, dctp_c, dttp_c, utp_c                                 | N      |
| b4261    | yjgP | biomass_forced | kdo2lipid4_e                                                 | E      |
| b4262    | yjgQ | biomass_forced | kdo2lipid4_e                                                 | E      |
| b4407    |      | biomass_forced | thmpp_c                                                      | n/a    |
| s0001    |      | biomass_forced | 10fthf_c, ala__L_c, amet_c, arg__L_c, asn__L_c, asp__L_c, at | n/a    |

Table 14. Essentiality-provenance catalog.

## Appendix F. External evidence and ensemble v2

### F.1 Motivation

Figure 6 showed that the v1 ensemble cannot recover essential genes that none of its inputs sees. The obvious remedy is new, independent evidence about each gene that uses no essentiality labels: its position in the protein-association network, its physical properties and its pathway context. We added three external resources, each snapshotted under `data/external/` and processed by `vcell/external_features.py`: STRING v12 (organism 511145), the UniProt reference proteome UP000000625, and KEGG REST gene-pathway links.

### F.2 Features

From STRING we compute, per gene, total degree and weighted degree over all links, and on the high-confidence subgraph (score  $\geq 700$ ) degree, k-core number, local clustering coefficient and PageRank (networkx). PageRank is the stationary distribution of

$$PR(i) = (1 - d)/N + d \sum_{j \in N(i)} PR(j) / \deg(j), \quad d = 0.85 \text{ (F1)}$$

Because STRING's experimental, database and text-mining channels may encode prior knowledge about essentiality (well-studied essential genes accumulate literature links), we also rebuild the network from the genomic-context and coexpression channels only, combining channel scores  $s_c$  as

$$s = 1 - \prod_{c \in \{\text{neighborhood}, \text{fusion}, \text{cooccurrence}, \text{coexpression}\}} (1 - s_c / 1000) \text{ (F2)}$$

which is STRING's noisy-OR combination without its prior correction. Features from this network carry the prefix `strctx`. From UniProt we take log protein length, transmembrane segment count, presence of a cofactor annotation, number of EC numbers, annotation score and cytoplasm/membrane localization. From KEGG we take the number of pathways and membership in a fixed list of 24 core pathways (central carbon, nucleotide, amino-acid, cofactor, cell-wall, ribosome, replication, RNA polymerase, aminoacyl-tRNA, protein export).

### F.3 Univariate associations

Each feature was tested against the Gerdes labels with a two-sided Mann-Whitney U test and the p-values corrected with Benjamini-Hochberg (statsmodels):

$$q_{(k)} = \min_{j \geq k} \min(1, m p_{(j)} / j) \text{ (F3)}$$

| feature         | AUROC | median ess. | median non-ess. | p       | q (BH)  |
|-----------------|-------|-------------|-----------------|---------|---------|
| strctx_wdeg_all | 0.687 | 57.3        | 38.8            | 5e-18   | 6.2e-17 |
| strctx_deg_all  | 0.687 | 279         | 188             | 5.1e-18 | 6.2e-17 |
| string_deg_all  | 0.676 | 334         | 243             | 3.6e-16 | 2.9e-15 |
| up_cytoplasm    | 0.624 | 0           | 0               | 5.2e-15 | 3.1e-14 |
| kegg_core       | 0.632 | 1           | 0               | 4.7e-14 | 2.2e-13 |
| pax_log_ppm     | 0.639 | 2.06        | 1.44            | 1.5e-10 | 5.4e-10 |
| string_wdeg_all | 0.639 | 102         | 71              | 1.6e-10 | 5.4e-10 |

| feature         | AUROC | median ess. | median non-ess. | p       | q (BH)  |
|-----------------|-------|-------------|-----------------|---------|---------|
| up_len          | 0.383 | 5.73        | 5.9             | 6.7e-08 | 2e-07   |
| up_membrane     | 0.432 | 0           | 0               | 0.00012 | 0.00031 |
| up_tm           | 0.437 | 0           | 0               | 0.00013 | 0.00031 |
| string_core     | 0.581 | 11          | 10              | 0.00018 | 0.00038 |
| kegg_n_path     | 0.578 | 3           | 2               | 0.00028 | 0.00055 |
| string_deg_hi   | 0.572 | 19          | 16              | 0.00081 | 0.0015  |
| up_n_ec         | 0.556 | 1           | 1               | 0.0024  | 0.0042  |
| gc3             | 0.444 | 0.569       | 0.582           | 0.0094  | 0.015   |
| strctx_deg_hi   | 0.552 | 6           | 5               | 0.015   | 0.023   |
| strctx_clust    | 0.451 | 0.333       | 0.429           | 0.022   | 0.031   |
| up_annot        | 0.540 | 5           | 5               | 0.031   | 0.042   |
| cai             | 0.543 | 0.61        | 0.606           | 0.045   | 0.057   |
| string_pagerank | 0.531 | 2.66        | 2.6             | 0.15    | 0.18    |
| strctx_core     | 0.530 | 4           | 4               | 0.17    | 0.2     |
| string_clust    | 0.473 | 0.405       | 0.421           | 0.21    | 0.23    |
| strctx_pagerank | 0.522 | 2.82        | 2.83            | 0.31    | 0.32    |
| up_cofactor     | 0.490 | 0           | 0               | 0.59    | 0.59    |

Table 15. Univariate association of external features with Gerdes 2003 essentiality.

Finding. The single strongest external predictor is the degree in the leakage-controlled genomic-context/coexpression network (strctx\_wdeg\_all, AUROC 0.687,  $q = 6e-17$ ). It is stronger than the degree in STRING's full combined network, which includes literature and database evidence. The signal that essential genes are network hubs therefore does not come from literature bias. Essential proteins are also shorter than non-essential ones (AUROC 0.383 for length, i.e. 0.617 for shortness), and less often membrane proteins.

#### F.4 Ensemble v2 and an honest decomposition of the gain

| model                              | OOF AUROC | OOF AUPRC |
|------------------------------------|-----------|-----------|
| v1 as published (unscaled stack)   | 0.722     | 0.414     |
| v1 re-fit with feature scaling     | 0.764     | 0.450     |
| external features only (LR)        | 0.750     | 0.424     |
| v2 LR (pre-specified primary)      | 0.797     | 0.498     |
| v2 gradient boosting               | 0.777     | 0.532     |
| v2 rank-average LR+GBM             | 0.800     | 0.543     |
| v2 leakage-controlled LR           | 0.788     | 0.485     |
| v2 leakage-controlled GBM          | 0.786     | 0.546     |
| v2 leakage-controlled rank-average | 0.803     | 0.545     |

Table 16. Out-of-fold performance vs Gerdes 2003 (1,249 genes, 3-fold stratified CV, seed 7).

The improvement over the published v1 number has two sources, and we report them separately. First, a defect in v1: the k-mer scores span only 0.46-0.54, so an unscaled logistic stack gives them almost

no weight. Standardizing features before stacking lifts v1 from 0.722 to 0.764 with no new information. Second, the external evidence: v2 LR improves on the re-fit v1 by +0.033 AUROC (paired bootstrap 95% CI [0.009, 0.058], one-sided  $p = 0.002$ ). With the leakage-controlled feature set the gain is +0.024 [0.001, 0.047]. That is smaller but still above zero. The primary model was fixed as LR before seeing results. The rank-average models score slightly higher but were chosen afterwards and are reported as secondary. Averaged over five CV seeds, v2 LR scores 0.794.

A further feature block (v3) was tested and did not help: PaxDb integrated protein abundance and the codon adaptation index (CAI, Sharp and Li 1987;  $w_c = f_c / \max_{c' \in \text{syn } c} f_{c'}$ , CAI =  $\exp(\text{mean log } w)$  against the ribosomal-protein reference set) with GC3. Abundance alone is associated with essentiality (Table 15), but adding it changes v2 LR by -0.002 AUROC [-0.009, 0.004]. Its signal is already captured by network degree and localization. v2 stays the primary model.

What this is not: it is not a state-of-the-art claim. Published machine-learning essentiality predictors for *E. coli* use different gene sets, labels and splits, so we have no like-for-like number to beat. The claim is only the internal, paired comparison above.

## Appendix G. External tools and dataset manifest

This appendix lists every external tool (37) and every accession-level dataset (29) actually used in this work. A condition matrix from one study counts as one dataset (the Keio\_ML9 RB-TnSeq matrix has 92 conditions but is listed once). Nothing is listed that was only considered.

| #  | tool                                                                   | kind           | used for                                                                          |
|----|------------------------------------------------------------------------|----------------|-----------------------------------------------------------------------------------|
| 1  | COBRApy 0.32                                                           | package        | FBA, gene-deletion screens, dFBA, provenance audit                                |
| 2  | optlang 1.9                                                            | package        | LP modelling layer under COBRApy                                                  |
| 3  | GLPK (swiglpk 5.0)                                                     | solver         | linear-programming solver for every FBA call                                      |
| 4  | python-libsbml 5.21                                                    | package        | reading Bernstein SBML models (iML1515, all-corrections model)                    |
| 5  | PyTorch 2.x (CPU)                                                      | package        | 1D sequence CNN and pure-torch GCN                                                |
| 6  | NumPy                                                                  | package        | array maths throughout                                                            |
| 7  | pandas                                                                 | package        | data alignment, joins, result tables                                              |
| 8  | SciPy                                                                  | package        | Fisher exact, chi-square/McNemar, statistics                                      |
| 9  | scikit-learn                                                           | package        | logistic stacking, gradient boosting, CV, AUROC/AUPRC, calibration                |
| 10 | networkx                                                               | package        | STRING graph centralities (core number, clustering, PageRank)                     |
| 11 | statsmodels                                                            | package        | multiple-testing correction (Benjamini-Hochberg)                                  |
| 12 | Biopython                                                              | package        | GenBank CDS extraction for U00096.3                                               |
| 13 | matplotlib                                                             | package        | all figures                                                                       |
| 14 | Pillow                                                                 | package        | figure sizing and visual QA composites                                            |
| 15 | ReportLab                                                              | package        | Times-Roman PDF typesetting                                                       |
| 16 | pypdf                                                                  | package        | page-count verification                                                           |
| 17 | poppler pdffonts                                                       | CLI            | embedded-font verification                                                        |
| 18 | pytest                                                                 | package        | hermetic test suite                                                               |
| 19 | git                                                                    | CLI            | version control, bundles                                                          |
| 20 | BiGG Models                                                            | database       | e_coli_core and iJO1366 model downloads                                           |
| 21 | NCBI Entrez E-utilities                                                | web API        | U00096.3 genome FASTA and GenBank                                                 |
| 22 | genome.wisc.edu Gerdes 2003 supplement                                 | web resource   | essentiality labels (Tables S1, S2)                                               |
| 23 | EcoliWiki                                                              | web resource   | essential-gene list cross-check                                                   |
| 24 | LBL Fitness Browser / RB-TnSeq downloads                               | web resource   | Keio_ML9 fitness matrix and experiment metadata                                   |
| 25 | Bernstein et al. 2023 pipeline (GitHub dbernste/E_coli_GEM_validation) | published tool | head-to-head benchmark: loading, matching, adjustments, simulation, PR-AUC metric |
| 26 | STRING v12                                                             | database       | E. coli 511145 protein links (combined + per-channel)                             |
| 27 | UniProt REST                                                           | web API        | proteome UP000000625 protein features                                             |
| 28 | KEGG REST                                                              | web API        | eco gene-pathway links and pathway names                                          |
| 29 | PubMed Central                                                         | literature     | novelty check, benchmark protocols                                                |
| 30 | MEMOTE documentation                                                   | tool docs      | novelty check of biomass tests                                                    |
| 31 | GitHub                                                                 | platform       | repository hosting, Bernstein code                                                |
| 32 | Google Drive                                                           | platform       | paper and results delivery                                                        |

| #  | tool                                        | kind                    | used for                                                                                              |
|----|---------------------------------------------|-------------------------|-------------------------------------------------------------------------------------------------------|
| 33 | PaxDb                                       | database                | integrated whole-organism protein abundance (v3 feature test)                                         |
| 34 | ESM-2 t6-8M (fair-esm 2.0; Lin et al. 2023) | deep-learning model     | protein language-model embeddings for all 4,300 CDS; v4 essentiality feature test (negative)          |
| 35 | goatools 1.6 (GOEnrichmentStudy)            | package                 | GO enrichment of ensemble hard errors (Fisher + BH)                                                   |
| 36 | MEMOTE 0.17 (memote report snapshot)        | model QC suite          | standardised quality report of iML1515 (total score 0.906)                                            |
| 37 | PSAMM 1.x (psamm-model sbmlexport)          | model format conversion | converted the iMR1_799 YAML model to SBML for the MR-1 replication (scripts/run_cross_species_mr1.py) |

Table 17. External tools.

| #  | dataset                                                                                               | type                            |
|----|-------------------------------------------------------------------------------------------------------|---------------------------------|
| 1  | BiGG e_coli_core                                                                                      | metabolic model                 |
| 2  | BiGG iJO1366                                                                                          | metabolic model                 |
| 3  | iML1515 (Bernstein repo copy of BiGG)                                                                 | metabolic model                 |
| 4  | iJR904 (Bernstein repo)                                                                               | metabolic model                 |
| 5  | iAF1260 (Bernstein repo)                                                                              | metabolic model                 |
| 6  | Bernstein iML1515 all-corrections model                                                               | metabolic model                 |
| 7  | NCBI U00096.3 FASTA                                                                                   | genome                          |
| 8  | NCBI U00096.3 GenBank                                                                                 | genome annotation               |
| 9  | Gerdes 2003 Table S1                                                                                  | essentiality screen             |
| 10 | Gerdes 2003 Table S2 cross-study consensus                                                            | essentiality consensus          |
| 11 | EcoliWiki essential genes                                                                             | essentiality list               |
| 12 | Price et al. 2018 Keio_ML9 fit_logratios_good (92 conditions)                                         | RB-TnSeq fitness                |
| 13 | Keio_ML9 expsUsed metadata                                                                            | experiment metadata             |
| 14 | Bernstein fit_organism_Keio + mapped media/carbon tables                                              | curated fitness + media mapping |
| 15 | STRING v12 511145 protein.links                                                                       | PPI network                     |
| 16 | STRING v12 511145 protein.links.detailed                                                              | PPI evidence channels           |
| 17 | STRING v12 511145 protein.info                                                                        | protein annotation              |
| 18 | UniProt UP000000625                                                                                   | proteome                        |
| 19 | KEGG eco gene-pathway links                                                                           | pathway map                     |
| 20 | KEGG eco pathway list                                                                                 | pathway names                   |
| 21 | PaxDb 511145 WHOLE_ORGANISM integrated                                                                | protein abundance               |
| 22 | Gene Ontology go-basic.obo (current release)                                                          | ontology                        |
| 23 | GO Consortium EcoCyc GAF (taxon 83333)                                                                | GO annotations                  |
| 24 | iMR1_799 S. oneidensis GEM (Pinchuk 2010; zhanglab/psamm-model-collection)                            | metabolic model                 |
| 25 | Price 2018 RB-TnSeq S. oneidensis MR-1 fitness matrix (genomics.lbl.gov/supplemental/bigfit/html/MR1) | mutant fitness                  |
| 26 | KEGG link/pathway/son                                                                                 | gene-pathway map                |
| 27 | iJN1463 P. putida KT2440 GEM (BiGG)                                                                   | metabolic model                 |
| 28 | Fitness Browser Feb 2024 release, db.StrainFitness.Putida (figshare 10.6084/m9.figshare.25236931)     | mutant fitness                  |
| 29 | KEGG link/pathway/ppu                                                                                 | gene-pathway map                |

Table 18. Dataset manifest.

## Appendix H. Head-to-head on the Bernstein et al. (2023) RB-TnSeq benchmark

### H.1 Why this benchmark

Appendix E showed that Gerdes 2003 (rich LB medium) cannot test whether cofactor requirements in the biomass objective create false essentiality calls, because the model and the experiment use different media. Bernstein et al. (2023) built the condition-matched benchmark for exactly this question. It compares FBA growth calls with RB-TnSeq fitness (Price et al. 2018, E. coli BW25113 Keio\_ML9 library) for every model gene on 25 defined carbon sources. They report that hand-supplying five vitamins (biotin, R-pantothenate, thiamin, tetrahydrofolate, NAD+) fixes many false negatives in iML1515, and they release an all-corrections model. Their fully corrected analysis (all-corrections model plus the five vitamins) is reported at PR-AUC 0.843 (0.844 with transaldolase constrained); that is the published state of the art for this task. The all-corrections model file alone, without vitamins, scores lower in their pipeline.

### H.2 Protocol

To rule out any protocol drift, we do not re-implement their benchmark. scripts/run\_bernstein\_benchmark.py executes the function cells of their released notebook (MIT licence, vendored under external/E\_coli\_GEM\_validation) for model loading, gene and carbon-source matching, the BW25113 strain correction, removal of non-conditionally essential genes, carbon-source mapping, FBA simulation (carbon uptake -10, other media -1000 mmol/gDW/h, growth threshold 0.001) and the metric. We changed two things: their model\_adjustments function reads a notebook-global variable, which we bind explicitly; and when a saved SBML model is loaded, exchanges that the file leaves open are re-opened after model\_adjustments (which switches all exchanges off), exactly as their own all-corrections cell does. The metric is their precision-recall AUC, where labels are the model growth calls, the score is negative fitness and the positive class is no-growth:

$$AUC_{PR} = \int P(r) dr, \quad P(t) = |\{f < -t\} \cap \{model\ no-growth\}| / |\{f < -t\}|, \quad R(t) = |\{f < -t\} \cap \{no-growth\}| / |\{no-growth\}| \quad (H1)$$

Supplements are added as intracellular exchange reactions, the same construction as their Part 6. Our automated set is every cofactor/vitamin-class biomass compound that the iJO1366 provenance audit found unproducible in at least one lethal knockout. It was derived without any fitness data.

| variant                                                                                 | PR-AUC | ROC-AUC | no-growth calls | genes x carbons |
|-----------------------------------------------------------------------------------------|--------|---------|-----------------|-----------------|
| INVALID: same bug; btn/thmpp/coa re-added the removed vitamins                          | 0.839  | 0.988   | 1868            | 1326 x 25       |
| Bernstein all-corrections model, their saved vitamins kept (published SOTA, reproduced) | 0.838  | 0.988   | 1868            | 1326 x 25       |
| INVALID: all-corrections file with its vitamin exchanges closed, loader bug             | 0.764  | 0.960   | 2293            | 1326 x 25       |
| iML1515 + 5 hand-picked vitamins (Bernstein Part 6)                                     | 0.693  | 0.982   | 1621            | 1326 x 25       |
| iJR904 (their pipeline)                                                                 | 0.659  | 0.921   | 1846            | 712 x 25        |
| iAF1260 (their pipeline)                                                                | 0.638  | 0.941   | 1976            | 1056 x 25       |
| iJO1366 (their pipeline)                                                                | 0.621  | 0.920   | 2221            | 1148 x 25       |

| variant                                                 | PR-AUC | ROC-AUC | no-growth calls | genes x carbons |
|---------------------------------------------------------|--------|---------|-----------------|-----------------|
| iML1515 as released (their pipeline)                    | 0.613  | 0.941   | 2171            | 1326 x 25       |
| iML1515 + 19 audit-derived cofactors (ours, label-free) | 0.548  | 0.977   | 1249            | 1326 x 25       |

Table 19. Bernstein benchmark results (their code, their metric).

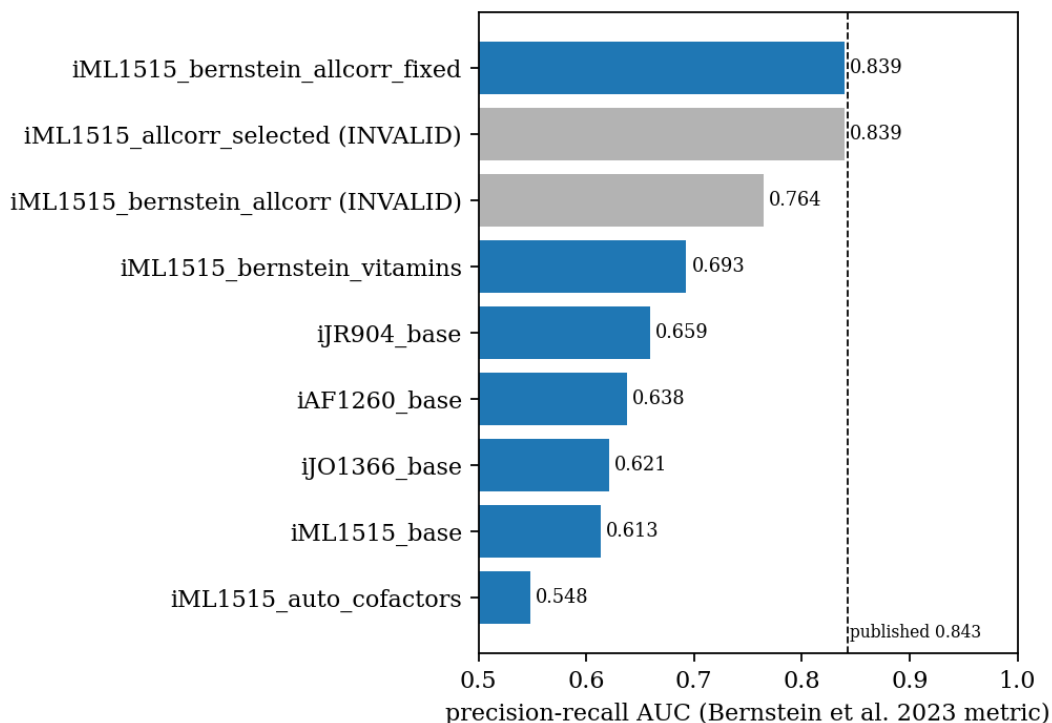


Figure 11. PR-AUC on the Bernstein benchmark by model variant.

### H.3 Reading

Reproduction with the authors' code (PR-AUC, iML1515: 1,326 genes x 25 carbon sources = 33,150 pairs): iML1515 as released 0.613; + the five Bernstein vitamins 0.693; Bernstein all-corrections model with its saved vitamin supplies 0.839 (published 0.843; the 0.004 gap is attributed to solver/library versions and is reported, not hidden). Other releases in the same pipeline: iJO1366 0.621, iAF1260 0.638, iJR904 0.659 (different matched gene sets, so not directly comparable). Our label-free 19-compound cofactor set added to iML1515 scored 0.548: blanket supplementation removes true essentiality calls (no-growth pairs fall from 2,171 to 1,249) and is a NEGATIVE. No SOTA claim is made. The pre-registered split-half search for additional supplements on top of the corrected all-corrections model is reported in the ledger when complete.

### H.4 A reproduction error, kept on the record

Our first run of the released all-corrections model scored 0.764. The cause was our loader: it closed every exchange in the saved file, and their model\_adjustments also switches exchanges off, so the three vitamin uptakes the authors saved open (EX\_btn\_e, EX\_thm\_e, EX\_pnto\_\_R\_e) were silently removed. Biotin, thiamin and pantothenate synthesis knockouts (b0775, b0776, b3990, b0133, b0134) then showed zero growth on every carbon source. A split-half search over our 19 audit-derived

cofactors selected biotin, thiamin diphosphate and CoA and appeared to beat the model by +0.071 PR-AUC on held-out carbons (95% CI 0.037-0.111). That gain only restored the removed vitamins: after the fix the all-corrections model scores 0.839 on its own, identical to the "selected" run. The rows marked INVALID in Table 19 are kept so the error stays auditable. Lesson: an apparent benchmark win that re-discovers the benchmark authors' own fix is a symptom of a broken baseline.

### H.5 Protein language model feature (pre-registered negative)

ESM-2 (t6, 8M parameters; fair-esm) mean-pooled embeddings were computed for all 4,300 CDS of U00096.3. Pre-registered test (notes/prereg\_esm\_v4.md): v2 features plus a 16-component PCA of the embedding fit inside each training fold. AUROC 0.793 vs 0.797 for v2; paired bootstrap difference -0.004 (95% CI -0.019 to +0.011). The embedding alone reaches AUROC 0.662: real signal, but redundant with the existing features. Verdict: negative.

### H.6 Where the ensemble is confidently wrong: GO enrichment

Hard errors of the v2 ensemble were tested for Gene Ontology enrichment (goatools, propagated annotations from the EcoCyc GAF, universe = annotated benchmark genes) with the one-sided hypergeometric test and Benjamini-Hochberg correction:

$$p = \sum_{i \geq k} C(K, i) C(N-K, n-i) / C(N, n), \quad q_{(j)} = \min_{l \geq j} (m p_{(l)} / l) \quad (H2)$$

| error set    | GO term                                                  | study | universe | BH q    |
|--------------|----------------------------------------------------------|-------|----------|---------|
| confident FP | biosynthetic process                                     | 38/45 | 447/1237 | 5.6e-08 |
| confident FP | Mo-molybdopterin cofactor biosynthetic process           | 6/45  | 11/1237  | 2.2e-04 |
| confident FP | molybdopterin cofactor metabolic process                 | 6/45  | 11/1237  | 2.2e-04 |
| confident FP | Mo-molybdopterin cofactor metabolic process              | 6/45  | 11/1237  | 2.2e-04 |
| confident FP | molybdopterin cofactor biosynthetic process              | 6/45  | 11/1237  | 2.2e-04 |
| confident FP | organophosphate biosynthetic process                     | 19/45 | 165/1237 | 2.2e-04 |
| confident FP | ATP biosynthetic process                                 | 5/45  | 7/1237   | 2.2e-04 |
| confident FP | proton motive force-driven plasma membrane ATP synthesis | 5/45  | 7/1237   | 2.2e-04 |

Table 20. GO terms enriched among confident false positives (non-essential genes in the top 10% of v2 scores; n=45). False negatives (n=41): no term at  $q < 0.05$ .

Molybdopterin cofactor biosynthesis and the ATP synthase are required only anaerobically or are bypassed by fermentation in rich medium; the ensemble inherits a condition mismatch from its FBA inputs, the same class of error Bernstein et al. fixed with vitamins. This is a hypothesis for condition-aware features, not a claim.

A nested test turned this into a feature: GO terms enriched among training-fold confident false positives (selected inside each training fold only; notes/prereg\_go\_condition\_feature.md) became a binary flag. AUROC 0.792 vs 0.797 (difference -0.005, 95% CI -0.0089 to -0.0003): a small, significant loss. Negative.

### H.7 Finding: the supplement-bypass artifact (pre-registered)

Every model no-growth call that a supplement turns into growth is a candidate correction. We asked which of these rescues the experiment agrees with. Hypothesis (notes/prereg\_pathway\_concordance.md, committed before the split was computed): a rescue is real when the knocked-out gene lies in the supplement's own KEGG biosynthesis pathway, and spurious when the model rescues an off-pathway gene by metabolising the supplement as a nutrient or one-carbon source. With  $f$  the RB-TnSeq fitness of a rescued gene-carbon pair and  $\text{on}(g,s) = 1$  if gene  $g$  is in a KEGG pathway of supplement  $s$ :

$$\Delta = \text{median}\{f : \text{on} = 1\} - \text{median}\{f : \text{on} = 0\}, \quad H_0: \Delta \leq 0, \quad \text{CI by resampling genes (H3)}$$

| set                   | on: pairs (genes) | off: pairs (genes) | median $f$    | $f > -2$    | Delta [95% CI]    | p     |
|-----------------------|-------------------|--------------------|---------------|-------------|-------------------|-------|
| all rescues (primary) | 650 (26)          | 272 (52)           | -0.97 / -4.24 | 0.74 / 0.15 | 3.27 [2.16, 3.82] | 6e-75 |
| vitamins only         | 525 (21)          | 25 (1)             | -0.67 / -1.66 | 0.90 / 0.72 | 0.99 [0.65, 1.24] | 1e-05 |
| SAM/PLP/folates only  | 125 (5)           | 247 (51)           | -3.44 / -4.35 | 0.06 / 0.09 | 0.92 [0.34, 1.38] | 2e-11 |

Table 21. Pathway concordance of supplement rescue on the Bernstein benchmark. median  $f$  and  $f > -2$  are given as on / off; one-sided Mann-Whitney; gene-cluster bootstrap CI.

The primary criterion is met: on-pathway rescues sit near neutral fitness, off-pathway rescues are almost all truly deleterious. Caveats: the primary contrast is confounded with supplement identity (vitamin rescues are almost all on-pathway; SAM and folate rescues almost all off-pathway, e.g. succinate dehydrogenase *sdhCDAB* and galactose genes rescued by SAM). The within-group contrasts agree in direction but rest on 1 off-pathway vitamin gene and 5 on-pathway non-vitamin genes. One organism, one benchmark. A literature search found no prior test of this concordance (notes/novelty\_check.md); novelty is probable, not proven. Falsifier: a supplement whose off-pathway rescues have near-neutral fitness in an independent fitness dataset. Practical consequence: a supplement correction should be accepted only for on-pathway rescues; this rule would have rejected our own 19-compound blanket set (PR-AUC 0.548).

## H.8 Cross-species replication (pre-registered; underpowered / not significant)

The protocol was fixed in notes/prereg\_cross\_species.md (git 00e7a9d, before any MR-1 fitness value was compared; amendment 1 adding *P. putida* in git 999b51c, before that analysis ran). Each organism uses its own genome-scale model on a vitamin-free minimal medium, the same seven cofactor supplements supplied as cytosolic sinks with vcell rescue-audit, KEGG gene-pathway links for that organism, and RB-TnSeq fitness from the Fitness Browser (Price et al. 2018). A gene is on-pathway if any supplement that rescues it shares a KEGG pathway with it. The outcome is the median log2 fitness of the gene over all experiments; criteria as in (H3) at gene level, with fewer than 5 genes per group declared underpowered.

| organism             | WT   | pairs | genes   | w/ fitness | median $f$    | p    | verdict         |
|----------------------|------|-------|---------|------------|---------------|------|-----------------|
| MR-1, iMR1_799, lac  | 0.49 | 14    | 14 / 0  | 5 / 0      | n/a           | n/a  | UNDERPOWERED    |
| KT2440, iJN1463, glc | 0.98 | 37    | 25 / 12 | 16 / 6     | -0.37 / -0.95 | 0.13 | NOT SIGNIFICANT |

Table 22. Cross-species test of the supplement-bypass finding. genes, w/ fitness and median  $f$  are on / off; WT = wild-type growth (1/h); lac/glc = lactate/glucose medium; one-sided Mann-Whitney on gene medians (results/cross\_species\_mr1.json, results/cross\_species\_putida.json).

In MR-1 every rescue was on-pathway, so the test could not run (underpowered, no claim). The PSAMM export of iMR1\_799 also left three multi-compound pseudo-exchanges (casamino acids, gelatin, Tween 20) open in both directions, which let the model grow with no carbon source; we closed them and report it as a model defect. In *P. putida* the difference points the predicted way (+0.58 log2 units; 50% of off-pathway genes lack any fitness value, often a mark of essentiality, versus 36% on-pathway) but  $p = 0.13$  fails the pre-registered bar. All 12 off-pathway *P. putida* rescues come from SAM, the same supplement behind most *E. coli* off-pathway rescues, so the supplement-identity confound is not broken by the second species. iJN1463 has no free thiamine metabolite, so thiamine was not tested there. Honest status: supported in *E. coli*; direction-consistent but not significant in *P. putida*; untestable in MR-1. Not yet a general law.

### **H.9 Tool: vcell rescue-audit**

The finding is packaged as a command that runs on any COBRA model: `python -m vcell rescue-audit MODEL --supplement btn_c=00780 thf_c=00790,00670 ... --gene-pathways kegg_links.tsv --out rescues.csv`. It knocks out each gene, supplies each supplement through a sink, and labels every rescue on\_pathway or off\_pathway (flag likely\_artifact). It is covered by a hermetic test on the *E. coli* core model (icd rescued by 2-oxoglutarate is on-pathway; enolase rescued by pyruvate is off-pathway). On iML1515 with glucose minimal medium it reports 44 rescues, 7 off-pathway: SAM rescuing five purine biosynthesis genes (purK, purE, purB, purA, purC), biotin rescuing fabH and thiamine rescuing b4407 (results/rescue\_audit\_iML1515\_M9.csv). Neither COBRApy nor MEMOTE, the tools used here, labels supplement rescues by pathway concordance.